

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



Le Xuan An

Gradient-Routed Alignment with Capacity-aware Experts
for Multimodal Understanding

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Ngành: Khoa học máy tính

Giảng viên hướng dẫn: Tiến sĩ Nguyễn Văn Sơn

Giảng viên đồng hướng dẫn: Phó Giáo sư Võ Đình Hiếu

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VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



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**Gradient-Routed Alignment with Capacity-aware Experts
for Multimodal Understanding**

BACHELOR'S THESIS

Major: Computer Science

Advisor: Dr. Nguyen Van Son

Co-Supervisor: Associate Professor Vo Dinh Hieu

HANOI - 2026

Lời cảm ơn

Assurance

I hereby affirm that this thesis is my own original work. I have not submitted it previously for any academic or professional award.

I completed this thesis through my own efforts, with guidance and support from Dr. Nguyen Van Son and Associate Professor Vo Dinh Hieu, in partial fulfillment of the graduation requirements. I confirm that all information, data and findings presented here are original and free from plagiarism.

I have acknowledged every source that I consulted and included each one in the references. I accept full responsibility for the accuracy and integrity of this thesis under the regulations of VNU University of Engineering and Technology.

Hanoi, Ngày 12 tháng 5 năm 2026

Student

Le Xuan An

Tóm tắt

Khóa luận này nghiên cứu bài toán chưng cất tri thức đa giáo viên không đồng nhất, trong đó tri thức được truyền từ một tập mô hình giáo viên VLM đóng băng sang một mô hình học sinh VLM. Bài toán này khó vì các mô hình giáo viên có thể thuộc những họ kiến trúc khác nhau, sử dụng bộ tách từ (tokenizer) khác nhau và không hữu ích như nhau trên mọi mẫu dữ liệu. Vì vậy, cách lấy trung bình cố định giữa các mô hình giáo viên thường không tận dụng tốt sự khác biệt về năng lực của từng giáo viên trên từng cặp ảnh và câu hỏi.

Khóa luận đề xuất phương pháp GRACE (Gradient-Routed Alignment with Capacity-aware Experts). Tên gọi này nhấn mạnh hai thành phần chính của phương pháp: định tuyến (routing) để chọn tập mô hình giáo viên phù hợp cho từng mẫu và căn chỉnh khác bộ tách từ thông qua hàm mất mát trie-Wasserstein trên cây byte dùng chung. Phương pháp xử lý ba vấn đề thực tế trong chưng cất đa giáo viên không đồng nhất: lệch bộ tách từ, khác biệt về thế mạnh giữa các mô hình giáo viên và sự bất ổn sau bước định tuyến. Cụ thể, GRACE kết hợp hàm mất mát trie-Wasserstein, cổng định tuyến giáo viên phụ thuộc vào trạng thái của mô hình học sinh, định tuyến thưa top- k và cơ chế tinh chỉnh trọng số dựa trên gradient.

Trong mỗi bước huấn luyện, cổng định tuyến trước hết chọn một tập nhỏ các mô hình giáo viên cho từng mẫu. Sau đó, GRACE so sánh gradient chưng cất tri thức (KD) của từng mô hình giáo viên với gradient CE có giám sát của mô hình học sinh, làm mượt điểm số bằng trung bình động hàm mũ (EMA) và tạo trọng số KD cuối cùng trong tập giáo viên đã được định tuyến. Cách làm này giúp mô hình học sinh nhận tín hiệu chưng cất từ các giáo viên phù hợp hơn với mục tiêu học có giám sát.

Thực nghiệm trên TextVQA, DocVQA và ChartQA cho thấy GRACE đạt kết quả cao hơn tinh chỉnh trực tiếp và các phương pháp cơ sở chưng cất được so sánh trong thiết lập báo cáo. Trong thực nghiệm, mô hình học sinh được khởi tạo từ SmolVLM-500M và SmolVLM-256M. Với thiết lập bốn mô hình giáo viên, GRACE đạt 69.38, 73.11 và 80.20 trên TextVQA, DocVQA và ChartQA khi dùng SmolVLM-500M làm mô hình học sinh. Các điểm số tương ứng với SmolVLM-256M là 60.73, 60.48 và 72.32. Đối với SmolVLM-500M, mức cải thiện so với mô hình gốc lần lượt là 22.26%, 22.83% và 27.71%. Phương pháp cũng thu hẹp 55.03% khoảng cách tới mô hình giáo viên trên TextVQA, 39.53% trên DocVQA và 55.34% trên ChartQA.

Từ khóa: mô hình thị giác-ngôn ngữ, chưng cất tri thức, chưng cất đa giáo viên, định tuyến hỗn hợp chuyên gia (mixture-of-experts routing), chưng cất Wasserstein khác bộ tách từ

Abstract

We study heterogeneous multi-teacher distillation for vision-language models (VLMs), where supervision is transferred from a pool of frozen teachers to a student model. The setting is difficult because teachers come from different model families, use different tokenizers and provide complementary supervision across samples, which makes fixed teacher averaging unreliable.

We present Gradient-Routed Alignment with Capacity-aware Experts (GRACE), a framework designed to address three failure modes in this setting: tokenizer mismatch, heterogeneous teacher specialization and unstable post-routing weighting. The frozen heterogeneous teachers are treated as tokenizer-specific experts. Teacher assignment is gradient-routed: after sparse top- k routing selects a candidate set, routed weights are refined according to the agreement between each teacher’s KD gradient and the supervised CE gradient. To the best of our knowledge, GRACE is the first distillation framework to use shared byte-trie optimal transport for heterogeneous multi-teacher VLM distillation. GRACE combines a trie-Wasserstein KD loss over a shared byte trie, sparse top- k teacher routing and gradient-agreement refinement. Routed teacher weights are then derived from smoothed agreement scores over the routed teacher set.

In TextVQA, DocVQA and ChartQA experiments with SmolVLM-500M and SmolVLM-256M as student models, GRACE improves over direct finetuning and all compared distillation baselines. In the four-teacher setting, GRACE reaches 69.38, 73.11 and 80.20 on SmolVLM-500M and 60.73, 60.48 and 72.32 on SmolVLM-256M. For SmolVLM-500M, this corresponds to gains of 22.26%, 22.83% and 27.71% over the original student. The method also closes 55.03% of the original teacher gap on TextVQA, 39.53% on DocVQA and 55.34% on ChartQA. These results demonstrate GRACE as an effective approach to heterogeneous multi-teacher VLM distillation on text-rich VQA tasks.

Keywords: vision-language model, knowledge distillation, multi-teacher distillation, sparse teacher routing, gradient agreement, cross-tokenizer Wasserstein distillation

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List of Abbreviations

Abbr.	Full Term	Description
ANLS	Average Normalized Levenshtein Similarity	Evaluation metric used for DocVQA.
CE	Cross-Entropy	Supervised answer-token loss used during training.
CKA	Centered Kernel Alignment	Representation similarity measure used to compare hidden layers.
ChartQA	Chart Question Answering	Benchmark for question answering over charts.
DocVQA	Document Visual Question Answering	Benchmark for question answering on document images.
EOS	End of Sequence	Special token that marks the end of a generated answer span.
GRACE	Gradient-Routed Alignment with Capacity-aware Experts	Proposed multi-teacher distillation method in this thesis.
SHM-CKA	Hidden-State Matching with CKA	Baseline that combines logits distillation with hidden-state CKA matching.
JS	Jensen-Shannon	Symmetric divergence measure between probability distributions.
KD	Knowledge Distillation	Transfer of supervision from teacher models to a student model.
KL	Kullback-Leibler	Divergence used to compare probability distributions.
LLM	Large Language Model	Language model with large parameter scale and strong generation ability.
MLP	Multi-Layer Perceptron	Feedforward projection block used in the teacher routing network.
MoE	Mixture of Experts	Sparse routing framework that motivates teacher selection.
MTKD	Multi-Teacher Knowledge Distillation	Distillation setting that uses more than one teacher model.

Abbr.	Full Term	Description
OCR	Optical Character Recognition	Extraction of text from images or scanned documents.
OT	Optimal Transport	Distribution-matching view used to motivate trie-Wasserstein KD.
TWD	Tree-Wasserstein Distance	Tree-structured optimal-transport distance used as background for the proposed loss.
TextVQA	Text Visual Question Answering	Benchmark for question answering on natural images containing text.
ULD	Universal Logit Distillation	Tokenizer-agnostic distillation baseline based on ranked probability distributions.
VLM	Vision-Language Model	Model that processes visual and textual inputs together.
VLM-Eval-Kit	Vision-Language Model Evaluation Kit	Toolkit used to run standardized multimodal evaluation.
VQA	Visual Question Answering	Task of answering questions grounded in visual inputs.

Chapter 1

Introduction

1.1 Background

Vision-language models (VLMs) have emerged as a **central direction** in modern AI because they integrate visual perception with language reasoning. By coupling powerful image encoders with autoregressive language decoders through learned multimodal connectors, these models can interpret visual content, read text embedded in images and reason across modalities in ways that neither vision systems nor language models alone can easily achieve [1–3]. This architectural synthesis has unlocked a **broad set of capabilities**, including extracting information from scanned documents, understanding visually rich scenes and grounding language instructions in physical environments [4–8].

Driven by neural scaling laws, the most capable VLMs are usually built on large language backbones trained with large multimodal corpora. This recipe reliably improves performance but also raises computational cost [9, 10]. As model size grows, the gap between what is achievable in research and what is deployable in practice widens accordingly. The **models that perform best are often the hardest to deploy** when latency, memory footprint, energy consumption and hardware constraints matter. In robotics and embodied AI, an agent must process visual observations, interpret language instructions and produce action decisions in real time, often on resource-constrained hardware [7, 8]. Similar constraints appear in mobile applications, document-processing pipelines and enterprise deployments, where users and operators need models that are **fast, affordable and secure**. This widening **gap between frontier capability and practical deployment** creates a central tension for applied VLM research.

Knowledge distillation offers a principled path through this tension by compressing the behavior of a large teacher model into a compact student. The student avoids training from scratch on the same massive corpora and learns from the teacher’s output distributions and internal representations, which carry richer information than hard labels alone [11, 12]. Distillation has been widely used to produce smaller models that **preserve much of the capability of their larger counterparts** and recent lightweight VLMs demonstrate that this direction is practical for latency-sensitive and cost-sensitive deployment [13–15].

However, distillation in the VLM setting cannot be treated as a straightforward compression problem. Existing approaches each address part of the challenge, but **none covers it fully**. Response-based distillation aligns teacher and student output distributions effectively when both share a tokenizer, but **breaks down** when teacher and student use **different vocabularies** [11, 12]. Transport-based methods relax this assumption by comparing probability mass over decoded strings, but they are usually formulated for single-teacher transfer and do not determine which teacher should supervise a given sample [16–18]. Representation matching through hidden states or similarity measures such as CKA provides a tokenizer-independent signal, but does not directly optimize answer likelihood and **cannot resolve conflicts between teachers** [19]. Multi-teacher distillation methods show the value of pooling several teachers, but fixed or globally learned weights can mix useful supervision with weak or conflicting signals [20–22].

The gap is most visible in text-rich visual reasoning tasks such as document question answering, chart interpretation and scene text understanding, which require optical character recognition, spatial layout understanding and multi-step inference [4–6]. Although recent open-source VLMs have narrowed the gap with proprietary systems on common benchmarks [23–25], smaller models continue to struggle on these harder tasks, where limited parameter capacity makes it difficult to capture fine visual details, document structure and numerical relations. Closing this gap requires a method that combines **tokenizer-agnostic** distribution matching, **sample-dependent** teacher selection and **gradient-aware conflict resolution** in a single framework. This is the central motivation of the thesis.

1.2 Problem Statement

Given a pool of frozen teacher VLMs and a trainable student VLM, the goal is to learn a distillation strategy that transfers task performance to the student while reducing deployment cost. The strategy must satisfy four requirements: it must remain valid when teachers and student differ in architecture, tokenizer and output vocabulary, assign teacher weights per sample and not globally, select only a sparse top- k subset of teachers per input and detect and suppress distillation gradients that conflict with the supervised objective. This thesis addresses all four requirements in a unified framework. This thesis proposes **Gradient-Routed Alignment with Capacity-aware Experts (GRACE)** as a method that satisfies these requirements. GRACE combines tokenizer-agnostic output matching, sparse sample-dependent routing and gradient-agreement refinement into a unified distillation objective for heterogeneous VLMs. Its design is evaluated through three research questions:

- **RQ1: Performance.** How does GRACE compare with direct fine-tuning, same-family logits KD, ULD and heterogeneous multi-teacher baselines on TextVQA, DocVQA and ChartQA benchmarks?
- **RQ2: Component Analysis.** How much do trie-Wasserstein KD, sparse top- k routing and gradient-agreement refinement contribute to the final performance?
- **RQ3: Efficiency.** What execution-time and computational overhead does GRACE introduce during training?

1.3 Thesis Contributions

This thesis presents three main technical contributions to the field of knowledge distillation in vision-language models.

A geometry-aware cross-tokenizer distillation objective. Trie-Wasserstein KD maps decoded teacher and student tokens into a shared UTF-8 byte trie and defines distillation loss as subtree-mass imbalances under the induced tree metric. Unlike methods that treat all token-string mismatches as equally costly, the objective penalizes distributional divergence in proportion to lexical distance. A sparse top- K support with a residual tail edge keeps computation tractable without discarding probability mass or constructing a full vocabulary-to-vocabulary transport matrix.

An objective-aware refinement mechanism for routed multi-teacher distillation. Gradient-agreement refinement is a post-routing step that measures cosine similarity between each routed teacher’s distillation gradient and the supervised cross-entropy gradient, then adjusts teacher weights by their degree of objective alignment. Sparse routing captures sample-level teacher relevance, while gradient-agreement refinement captures compatibility with the supervised objective. The two mechanisms are complementary and operate in sequence.

Empirical evidence supporting output-level over representation-level matching in heterogeneous VLM distillation. Centered kernel alignment is tokenizer-agnostic and a natural baseline for heterogeneous distillation, but controlled ablations show it does not yield consistent improvements when used alone, even when teachers and students share no vocabulary. The combination of Trie-Wasserstein matching, sparse routing and gradient-agreement refinement outperforms both CKA matching and uniform teacher averaging across DocVQA, ChartQA and related benchmarks, indicating that tokenizer-aware output-level distillation is a more reliable basis for heterogeneous knowledge transfer.

1.4 Thesis Organization

The remainder of this thesis is structured as follows. Chapter 2 provides the background and related work necessary to situate GRACE, covering VLMs, knowledge distillation, tokenizer-mismatched distribution matching, sparse routing mechanisms and gradient conflict resolution. Chapter 3 presents GRACE in full. It begins with the trie-Wasserstein KD objective, which enables tokenizer-agnostic distribution matching, then introduces the sparse teacher router for sample-dependent teacher selection and concludes with the gradient-agreement refinement step and the combined training objective. Chapter 4 describes the experimental setup in detail, including the text-rich VQA datasets, student and teacher model configurations, baseline methods, evaluation metrics and implementation details. Chapter 5 reports results across the three research questions: main benchmark comparisons against fine-tuning and distillation baselines, component ablations that isolate the contribution of each GRACE module and an efficiency analysis of the routing and gradient-refinement overhead. Finally, the thesis concludes with a summary of the main findings and a discussion of promising directions for future work in heterogeneous multi-teacher VLM distillation.

Chapter 2

Background and Related Work

This chapter introduces the background concepts and related work used throughout the thesis. It first covers the building blocks of VLMs and knowledge distillation, then turns to the cross-tokenizer, routing and multi-teacher challenges. The final section connects these threads into the research gap addressed by this thesis.

2.1 Vision-Language Models

2.1.1 Architecture Families

Modern vision-language models often combine a vision encoder with a language model. On the visual side, most systems use a Vision Transformer pretrained with a contrastive objective, commonly CLIP [26] or SigLIP [27], which converts an image into a sequence of patch embeddings that capture both local and global visual features. A connector module then fuses these visual tokens with text, using cross-attention, a Q-Former or a learned projector [1–3]. This design separates visual perception from language generation: the vision encoder produces a compact sequence of image features, while the language model provides the autoregressive prior used to answer questions. Different VLM families mainly differ in where and how fusion happens, as shown in Figure 2.1. Dual-encoder models align image and text embeddings contrastively, cross-attention models let language tokens attend to visual features directly and projector-based models map visual embeddings into the language model hidden space and then decode with a standard causal language model. This thesis focuses on the projector-based generative design, because distillation targets output token distributions, not retrieval embeddings.

²Panels adapted from CLIP [26], Flamingo [1], BLIP-2 [2] and LLaVA [3].

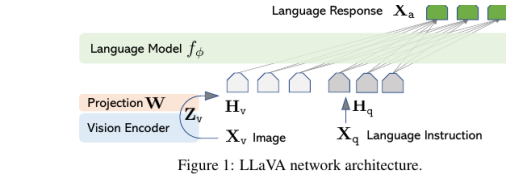
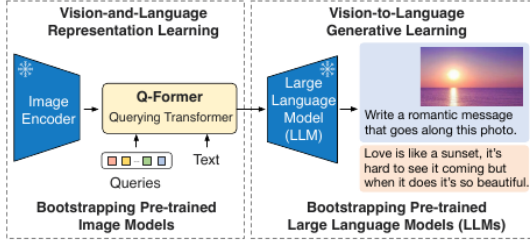
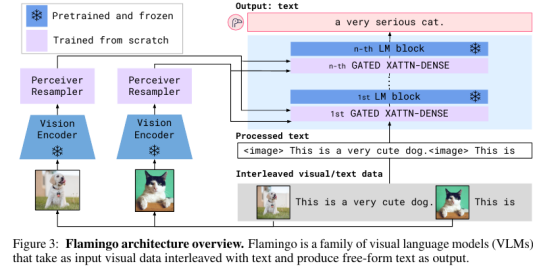
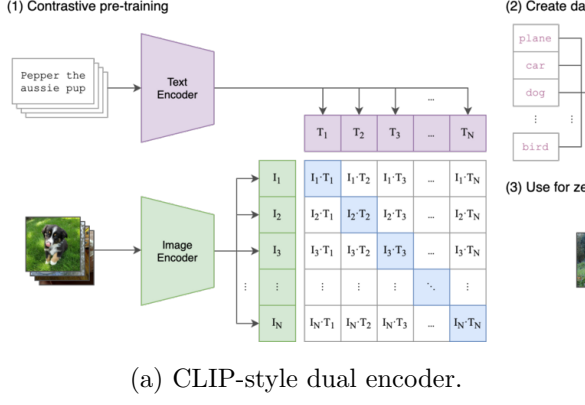


Figure 2.1: Representative architecture families of modern vision-language models.²

2.1.2 Autoregressive Answer Generation

The distillation setting in this thesis assumes a generative VLM student that produces answer tokens autoregressively. For language modeling, the Transformer maps each hidden state $\mathbf{h}_j$ to next-token logits and probabilities [28]:

$$\mathbf{z}_j = \mathbf{W}_o \mathbf{h}_j + \mathbf{b}_o, \quad p_j(v) = \text{softmax}(\mathbf{z}_j)_v = \frac{\exp(z_{j,v})}{\sum_{u \in \mathcal{V}} \exp(z_{j,u})}. \quad (2.1)$$

The full VLM can be abstracted as a mapping from multimodal input to output logits:

$$\mathbf{z} = f_\theta(x),$$

where x is a multimodal input and $\mathbf{z}$ denotes the output logits used for autoregressive answer generation. For an answer sequence $y = (y_1, \dots, y_L)$, the model factorizes the conditional probability as

$$p_\theta(y \mid x) = \prod_{j=1}^L p_\theta(y_j \mid x, y_{<j}). \quad (2.2)$$

Token-level distillation can then compare teacher and student distributions, such as $p_{t,j}$ and $p_{s,j}$, on the answer positions used for supervision [11, 28]. The loss is therefore naturally token-wise. At each answer position, the student distribution can be compared with either a hard label or a teacher distribution. This is important for VQA because short answers such as numbers, names and document fields may differ by only one or two tokens, so supervision must preserve position-level information, not only sequence-level accuracy after decoding the full answer.

2.2 Knowledge Distillation

2.2.1 Teacher-Student Objective

Knowledge distillation transfers information from a stronger teacher model to a student model [11, 12]. Let $\mathcal{L}_{CE}$ denote the supervised cross-entropy loss of the student. A generic distillation objective takes the form

$$\mathcal{L} = \lambda_{CE}\mathcal{L}_{CE} + \lambda_{KD}\mathcal{L}_{KD}, \quad (2.3)$$

where $\mathcal{L}_{KD}$ is a teacher-guided loss. In token-level distillation, the teacher and student are compared on their output distributions, not only on hard labels. The student therefore learns richer distributional structure, the relative probabilities over incorrect classes often called *dark knowledge*, that hard labels discard [11]. For a single answer position, a common same-vocabulary form is

$$\mathcal{L}_{KD} = \tau^2 \text{KL}\left(\mathbf{p}_t^{(\tau)} \parallel \mathbf{p}_s^{(\tau)}\right), \quad (2.4)$$

where τ is the distillation temperature. The cross-entropy term anchors the student to the ground-truth answer, while the KD term transfers the teacher’s relative preference over alternative tokens. In the heterogeneous setting of this thesis, this template remains useful, but the comparison term must be changed because teachers and students may not share the same tokenizer.

2.2.2 Temperature Scaling and Soft Targets

Let $\mathbf{z}_s$ and $\mathbf{z}_t$ denote student and teacher logits. With temperatures τ_s and τ_t , the corresponding distributions are

$$\mathbf{p}_s = \text{softmax}\left(\frac{\mathbf{z}_s}{\tau_s}\right), \quad \mathbf{p}_t = \text{softmax}\left(\frac{\mathbf{z}_t}{\tau_t}\right). \quad (2.5)$$

Many classical KD methods compare $\mathbf{p}_s$ and $\mathbf{p}_t$ using divergences such as forward KL or Jensen-Shannon divergence. However, such approaches usually assume that teacher and student share the same vocabulary [11, 16, 29]. The temperature controls how much probability mass is exposed beyond the top answer token. When $\tau > 1$, the softmax distribution becomes flatter and low-rank alternatives receive non-negligible probability. When $\tau < 1$, the distribution becomes sharper and approaches hard-label supervision. This matters because the teacher may encode useful semantic alternatives in the tail of the distribution, for example related words, number formats or subword variants. The same softening idea is used later before applying tokenizer-independent distribution matching. In practice, the student and teacher temperatures may be chosen separately. Using τ_t for the teacher controls how much dark knowledge is exposed, whereas τ_s controls the smoothness of the student distribution being matched. For this reason, the temperatures are treated as part of the distillation interface throughout this thesis, not merely numerical constants inside the softmax.

2.2.3 Answer-Token Supervision

For multimodal question answering, distillation should focus on *answer tokens*, not *prompt tokens* [3]. The KD loss is therefore applied only on supervised answer positions after masking ignored tokens and truncating student and teacher answer spans to a common length. The result is a token-aligned distillation signal that remains compatible with different teacher and student tokenizers. Let Ω_b^{CE} denote the non-ignored student answer positions for sample b , so the supervised loss is

$$\mathcal{L}_{CE}^{(b)} = \frac{1}{|\Omega_b^{CE}|} \sum_{j \in \Omega_b^{CE}} -\log p_s(y_j \mid x_b, y_{<j}). \quad (2.6)$$

The same masking principle is used for KD positions. This prevents the distillation objective from spending capacity on prompt reconstruction, padding tokens or formatting artifacts and concentrates the teacher signal on the answer span that is scored by the evaluation benchmarks. For multiple teachers, the KD position set can be teacher-specific because teacher tokenizers and generated answer spans may differ. The training code therefore constructs $\Omega_{b,i}^{KD}$ separately for each teacher T_i , then averages only over valid aligned answer positions for that teacher.

2.2.4 Response-Based vs. Feature-Based Distillation

Knowledge distillation methods can be grouped by the type of teacher signal they transfer. Common response-level divergences and feature-level similarity losses used in knowledge distillation are collected in Table 2.1 [11, 29]. In the table, p and q denote teacher and student output distributions over token events t . $m = \frac{1}{2}(p + q)$ is the JS midpoint distribution. $\mathbf{u} = \Phi_T(f_T(x, y))$ and $\mathbf{v} = \Phi_S(f_S(x, y))$ are teacher and student feature vectors. The index r marks feature dimensions or normalized feature bins. *Response-based* distillation uses the teacher’s output distribution or generated response as supervision. In its standard form, the loss compares next-token distributions:

$$\mathcal{L}_{resp} = \frac{1}{|\Omega|} \sum_{j \in \Omega} \text{KL}(\mathbf{p}_{t,j}^{(\tau)} \parallel \mathbf{p}_{s,j}^{(\tau)}), \quad (2.7)$$

where Ω is the set of supervised answer positions. This objective is closely tied to the student’s final prediction layer and therefore provides a task-facing signal. It is the classical form of KD introduced by Hinton et al. [11] and remains common in language-model and VLM compression [12, 30].

Feature-based distillation aligns intermediate representations between student and teacher layers. Given student and teacher hidden states $\mathbf{H}_s^{(\ell)}$ and $\mathbf{H}_t^{(m)}$ at selected layers, a direct matching objective is

$$\mathcal{L}_{feat} = \left\| \phi_s(\mathbf{H}_s^{(\ell)}) - \phi_t(\mathbf{H}_t^{(m)}) \right\|_2^2, \quad (2.8)$$

where ϕ_s and ϕ_t are projection functions that make the hidden states comparable. Feature-based methods are useful when output vocabularies are incompatible, because the comparison can be defined in representation space without vocabulary-space matching [19, 31]. However, they are less directly connected to answer likelihood and may transfer teacher-specific internal structure that is not useful for the downstream task. This distinction motivates treating response-level and representation-level signals as complementary, not interchangeable.

Table 2.1: Summary of response-level and feature-level KD objectives.

Response-Level Divergences		Feature-Level Losses	
Type	$D(p, q)$	Function $\mathcal{L}_F$	Expression
Forward KLD	$\sum_t p(t) \log \frac{p(t)}{q(t)}$	L2-Norm Distance	$\ \mathbf{u} - \mathbf{v}\ _2$
Reverse KLD	$\sum_t q(t) \log \frac{q(t)}{p(t)}$	L1-Norm Distance	$\ \mathbf{u} - \mathbf{v}\ _1$
JS Divergence	$\frac{1}{2} \text{KL}(p \parallel m) + \frac{1}{2} \text{KL}(q \parallel m)$	Cross-Entropy Loss	$-\sum_r \mathbf{u}_r \log \mathbf{v}_r$

2.2.5 Limitations of Same-Vocabulary Assumptions

Same-vocabulary KD assumes that teacher and student probabilities are indexed by the same token set $\mathcal{V}$. Under this assumption, the KL term

$$\text{KL}(\mathbf{p}_t \parallel \mathbf{p}_s) = \sum_{v \in \mathcal{V}} p_t(v) \log \frac{p_t(v)}{p_s(v)} \quad (2.9)$$

is well defined because token v has the same meaning in both distributions. Heterogeneous VLM distillation violates this assumption. A concrete example illustrates the problem. A teacher may tokenize the string “invoice” as one token while the student may split it into several subword or character-level tokens. The indices of $\mathbf{p}_t$ and $\mathbf{p}_s$ therefore refer to different discrete events, so a token-index KL divergence compares unrelated bins. This is the central obstacle addressed by cross-tokenizer and tokenizer-agnostic distillation methods [16].

2.3 Tokenizer-Agnostic Distribution Matching

2.3.1 The Vocabulary Mismatch Problem

Let $\mathcal{V}_t$ and $\mathcal{V}_s$ be the teacher and student vocabularies. In heterogeneous KD, the probability distributions $\mu_t \in \Delta^{|\mathcal{V}_t|-1}$ and $\mu_s \in \Delta^{|\mathcal{V}_s|-1}$ live on different supports. A direct same-index comparison is therefore invalid unless a shared alignment map is available. One possible abstraction is to decode each token into a string and compare probability mass over decoded strings:

$$\psi_t : \mathcal{V}_t \rightarrow \mathcal{S}, \quad \psi_s : \mathcal{V}_s \rightarrow \mathcal{S}, \quad (2.10)$$

where $\mathcal{S}$ denotes a shared space of strings. After decoding, distribution matching can be defined by how similar two decoded token strings are, not by whether their vocabulary indices are equal. This motivates transport-based and string-structured objectives, which replace same-vocabulary KL with a distance over probability mass on a shared semantic or lexical support.

2.3.2 Wasserstein Distance

When teacher and student vocabularies differ, standard divergences defined on a shared output space are not directly applicable. A distribution-matching framework that encodes the geometry of the output space is optimal transport. Optimal transport compares two probability measures by the minimum cost required to move mass from one distribution to the other [32]. For probability measures μ_s and μ_t , the p -Wasserstein cost can be written as

$$W_p^p(\mu_s, \mu_t) = \min_{\gamma \in \Pi(\mu_s, \mu_t)} \int_{\Omega_s \times \Omega_t} \|x - y\|^p d\gamma(x, y), \quad (2.11)$$

where $\Pi(\mu_s, \mu_t)$ is the set of couplings whose marginals are μ_s and μ_t . The coupling γ specifies how probability mass from the source distribution is transported to the target distribution. For discrete source weights $\mathbf{a}$, target weights $\mathbf{b}$ and transport cost matrix $\mathbf{C}$, the corresponding optimization problem can be written as

$$W(\mathbf{a}, \mathbf{b}) = \min_{\mathbf{P} \in \mathcal{U}(\mathbf{a}, \mathbf{b})} \langle \mathbf{P}, \mathbf{C} \rangle, \quad \mathcal{U}(\mathbf{a}, \mathbf{b}) = \{ \mathbf{P} \geq 0 \mid \mathbf{P}\mathbf{1} = \mathbf{a}, \mathbf{P}^\top \mathbf{1} = \mathbf{b} \}. \quad (2.12)$$

Here, $\mathbf{P}$ specifies the amount of mass moved from each source bin to each target bin in the transport plan. A key advantage of Wasserstein distances over pointwise divergences is geometric sensitivity. If two target distributions move farther away from a source distribution, a pointwise divergence may saturate once the supports no longer overlap. In contrast, Wasserstein distances continue to reflect the displacement cost. The setup in Figure 2.2 compares a source distribution with target distributions that shift progressively farther away. Pointwise divergences can saturate when supports stop overlapping, while Wasserstein distances continue to reflect displacement through the transport cost.

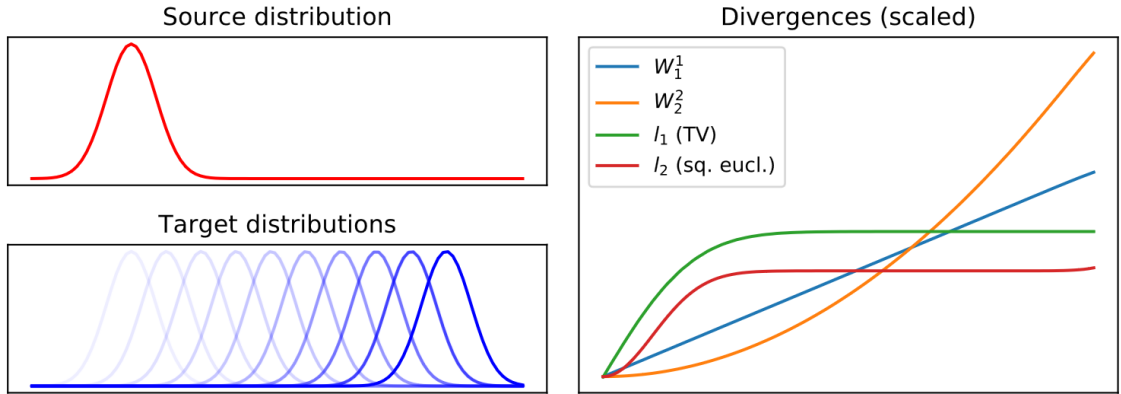


Figure 2.2: Geometric behavior of Wasserstein distance.¹

¹Adapted from Flamary and Courty [33].

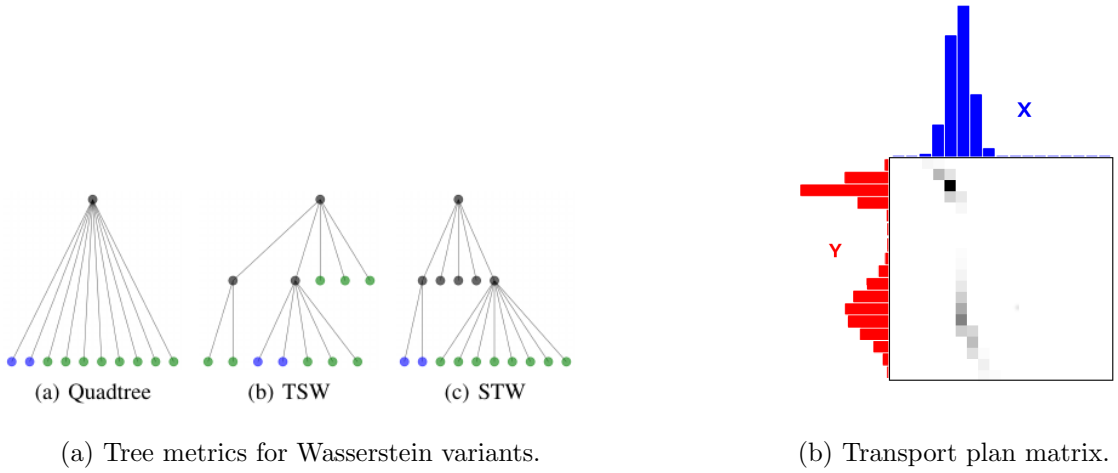


Figure 2.3: Discrete optimal transport views.¹

2.3.3 Tree-Structured Transport

Dense OT requires a pairwise ground cost between all source and target bins. In tokenizer-mismatched KD, however, token identity matching is not feasible because teacher and student vocabularies differ. A *tree metric* provides a structured alternative: probability mass is placed on a shared hierarchy and the distance is computed through *subtree-mass imbalance* [16, 32]. For W_1 on a tree metric, *tree-Wasserstein* distances make that idea explicit by placing probability mass on the tree and summing weighted subtree imbalances [17, 32]. For a rooted tree $T = (V, E)$ with edge weights $w(e)$, one convenient form is

$$W_T(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|, \quad (2.13)$$

where $\Gamma(e)$ denotes the subtree below edge e . The key point is that, on a tree metric, transport decomposes into edgewise subtree-mass imbalance and avoids a dense source-target coupling [17, 18]. The computation is efficient because the comparison follows the tree hierarchy directly and avoids materializing a full pairwise transport matrix. Figures 2.3a and 2.3b show the two commonly used OT views: a tree-metric view for fast subtree-mass comparisons and a matrix view for discrete transport plans. Tree-sliced OT extends this idea by averaging Wasserstein distances over tree metrics and shows that tree metrics preserve the geometric view of OT while admitting fast closed-form computations [18].

¹Panel (a) is adapted from Takezawa et al. [17]. Panel (b) is adapted from Wikimedia Commons and Peyré and Cuturi [32, 34].

2.4 Representation Matching

2.4.1 Hidden-State Alignment

Representation matching transfers knowledge through intermediate hidden-state comparison, not final token distributions. Let $\mathbf{H}_s \in \mathbb{R}^{n_s \times d_s}$ and $\mathbf{H}_t \in \mathbb{R}^{n_t \times d_t}$ denote student and teacher hidden-state matrices. Because the sequence lengths and dimensions may differ, alignment usually requires pooling, projection, interpolation or layer selection:

$$\tilde{\mathbf{H}}_s = A_s(\mathbf{H}_s), \quad \tilde{\mathbf{H}}_t = A_t(\mathbf{H}_t), \quad \mathcal{L}_{hid} = \left\| \tilde{\mathbf{H}}_s - \tilde{\mathbf{H}}_t \right\|_F^2. \quad (2.14)$$

Here, A_s and A_t denote alignment operations that place student and teacher activations in a comparable space. This family of methods is naturally appealing for heterogeneous models because it avoids direct vocabulary comparison and can use information from image tokens, text tokens or fused multimodal states [31].

2.4.2 Centered Kernel Alignment (CKA)

Centered Kernel Alignment (CKA) compares representations through their similarity structure, not through elementwise hidden-state equality [19]. Given two activation matrices $\mathbf{X} \in \mathbb{R}^{n \times d_x}$ and $\mathbf{Y} \in \mathbb{R}^{n \times d_y}$ for the same n examples, the linear CKA score can be written as

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{Y}^\top \mathbf{X}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \|\mathbf{Y}^\top \mathbf{Y}\|_F}. \quad (2.15)$$

Higher CKA values indicate that the two representations induce similar geometry over the examples. Because CKA is invariant to isotropic scaling and orthogonal transformations, it is often more stable than direct hidden-state matching when the teacher and student have different hidden dimensions. In this thesis, CKA is therefore relevant both as a possible representation-level distillation signal and as an analysis tool for measuring how closely student features follow teacher features.

2.4.3 Limitations of Representation-Only Matching

Representation matching does not require a shared tokenizer, but it also does not directly optimize answer probabilities. A hidden-state objective can make the student resemble the teacher internally while still leaving the final answer distribution poorly calibrated. In a multi-teacher setting, the problem is sharper: two teachers may have useful but incompatible internal representations and a representation-only objective does not specify which teacher should dominate for a given input. It also does not detect whether a teacher’s representation loss produces a gradient that agrees with the supervised answer loss. For these reasons, representation matching is best viewed as a complementary signal or diagnostic baseline, not a complete solution to heterogeneous multi-teacher VLM distillation.

2.5 Sparse Routing and Expert Selection

2.5.1 Mixture-of-Experts and Top- k Selection

Recent sparse expert models route each sample or token through only a subset of available experts, improving efficiency and allowing specialized computation [35–37]. Let $\mathbf{h}^{(b)}$ denote a pooled representation for sample b . A router network $G_\phi(\cdot)$ produces gate logits

$$\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)}), \quad \tilde{\mathbf{r}}^{(b)} = \text{AdjustedScores}(\mathbf{r}^{(b)}), \quad \mathbf{p}^{(b)} = \text{softmax}(\tilde{\mathbf{r}}^{(b)}), \quad (2.16)$$

where $p_i^{(b)}$ is the initial routing weight for expert i . The adjusted scores $\tilde{\mathbf{r}}^{(b)}$ may include expert bias, training noise or temperature scaling before the routing softmax. *Top- k routing* retains only the highest-scoring experts and zeros out all others. *Capacity constraints* prevent one expert from receiving too many assignments. If $\mathcal{S}_b = \text{TopK}(\mathbf{p}^{(b)}, k)$ is the selected expert set, the sparse expert output can be written as

$$\mathbf{y}^{(b)} = \sum_{i \in \mathcal{S}_b} \alpha_i^{(b)} E_i(\mathbf{h}^{(b)}), \quad \alpha_i^{(b)} = \frac{p_i^{(b)}}{\sum_{j \in \mathcal{S}_b} p_j^{(b)}}. \quad (2.17)$$

Only the selected experts are evaluated, so the activated computation can remain small even when the total number of expert parameters is large. The same abstraction is useful for teacher routing: each teacher can be treated as an expert that contributes a weighted distillation signal for a subset of samples.

2.5.2 Load Balancing Objectives

Load balancing prevents routing collapse onto a small subset of experts. In a sparse MoE layer, collapse is not only a modeling problem but also a systems problem: the overloaded experts determine the step latency, while under-used experts receive few gradients and remain poorly specialized [35, 38, 39]. Let a minibatch contain T routed items and let N denote the number of experts. For item b , the router produces probabilities $\mathbf{p}^{(b)} \in \Delta^{N-1}$. A top- k router induces the binary assignment mask

$$\mathbf{a}_i^{(b)} = \mathbf{1}\{i \in \text{TopK}(\mathbf{p}^{(b)}, k)\}, \quad n_i = \sum_{b=1}^T \mathbf{a}_i^{(b)}, \quad (2.18)$$

where n_i is the number of routed items assigned to expert i . Balanced routing corresponds to $n_i \approx kT/N$. When n_i is much larger than this value for a few experts, the router has effectively reduced the model to a small ensemble of busy experts while wasting the capacity of the remaining ones. The routing statistics used by many auxiliary losses can be written as a *soft* load and a *hard* load:

$$\bar{p}_i = \frac{1}{T} \sum_{b=1}^T p_i^{(b)}, \quad \bar{q}_i = \frac{1}{kT} \sum_{b=1}^T \mathbf{a}_i^{(b)}. \quad (2.19)$$

Here, $\bar{p}_i$ measures how much probability mass the router gives to expert i before hard selection, whereas $\bar{q}_i$ measures how often the expert is actually selected after sparse routing. The Switch Transformer balancing loss uses their scaled dot product [36]:

$$\mathcal{L}_{bal} = \alpha_{bal} N \sum_{i=1}^N \bar{p}_i \bar{q}_i. \quad (2.20)$$

The factor N keeps the magnitude approximately independent of the number of experts because the uniform solution gives $\sum_i (1/N)(1/N) = 1/N$. The hard load $\bar{q}_i$ is not differentiable through the top- k decision, but $\bar{p}_i$ is differentiable. The loss therefore gives the router a gradient signal that discourages assigning high average probability to already busy experts. Earlier sparsely-gated MoE work used separate penalties for expert *importance* and expert *load* [35]. If $\mathbf{I}$ is the vector of batchwise gate sums and $\mathbf{L}$ is a smooth estimate of the number of examples assigned to each expert, the penalty can be expressed through the squared coefficient of variation:

$$\text{CV}(\mathbf{z}) = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (z_i - \bar{z})^2}}{\bar{z}}, \quad \mathcal{L}_{imp} = \lambda_{imp} \text{CV}(\mathbf{I})^2, \quad \mathcal{L}_{load} = \lambda_{load} \text{CV}(\mathbf{L})^2. \quad (2.21)$$

This view makes the objective explicit: a balanced router should have low dispersion across experts. Later top-1 and top-2 routers often prefer the simpler dot-product form in Equation 2.20, combined with a capacity constraint, because it is easy to compute at scale [36, 38].

2.5.3 Capacity and Expert-Choice Routing

Expert capacity is the hard counterpart to the soft balancing objective. With a capacity factor γ , each expert is allowed to process at most

$$C = \left\lceil \gamma \frac{kT}{N} \right\rceil \quad (2.22)$$

routed items. A larger γ reduces overflow, but it also reserves more empty expert slots when the actual load is already close to uniform [36]. A smaller γ is more efficient but makes imbalance costly because over-capacity items must be dropped, rerouted or passed through a residual path. Thus, load balancing and capacity are coupled: good balancing allows a small capacity factor without losing many routed items. Load balancing can also be enforced by changing the routing direction. Expert-choice routing reverses the assignment: each expert selects a fixed number of items, while item-choice routing lets each item independently select experts. If c is a capacity factor, each expert receives

$$C_{EC} = \left\lfloor \frac{cT}{N} \right\rfloor \quad (2.23)$$

items by construction, so the expert load is balanced without an auxiliary loss. The tradeoff is that the number of experts assigned to each item becomes variable, which can be useful for heterogeneous inputs but less convenient for some autoregressive serving regimes [39]. In distributed expert-parallel systems, balancing individual experts may still be insufficient if experts are unevenly placed across devices. If $\mathcal{E}_d$ is the set of experts on device d , device-level loads can be summarized as

$$\bar{p}'_d = \sum_{i \in \mathcal{E}_d} \bar{p}_i, \quad \bar{q}'_d = \sum_{i \in \mathcal{E}_d} \bar{q}_i, \quad \mathcal{L}_{dev} = \alpha_{dev} D \sum_{d=1}^D \bar{p}'_d \bar{q}'_d. \quad (2.24)$$

DeepSeek-V2 uses this type of idea beyond expert-level balancing, adding device-level and communication-level balance terms so that computation and all-to-all traffic remain better distributed during expert-parallel training [37]. For teacher-routing distillation, the same mathematics applies with teachers playing the role of experts. After top- k selection, routed teacher weights can be renormalized as

$$\omega_i^{(b)} = \frac{\mathbf{a}_i^{(b)} p_i^{(b)}}{\sum_{j=1}^N \mathbf{a}_j^{(b)} p_j^{(b)} + \epsilon}. \quad (2.25)$$

Using these routed weights, the objective adds load balancing to the primary student loss:

$$\mathcal{L} = \mathcal{L}_{main} + \lambda_{bal} \mathcal{L}_{bal}. \quad (2.26)$$

The coefficient λ_{bal} must be chosen carefully. If it is too small, the router may collapse onto one dominant teacher. If it is too large, it can force nearly uniform routing and weaken useful specialization. In this thesis, load balancing is therefore best understood as a stabilizer for sparse teacher selection, not a replacement for the task or distillation loss.

2.5.4 Sample-Adaptive Routing

Sparse routing is useful because the best expert can depend on the input. In a generic sample-adaptive router, the selected expert set is a function of a sample representation:

$$\mathcal{S}_b = \text{TopK}(\text{softmax}(G_\phi(\mathbf{h}^{(b)})), k). \quad (2.27)$$

The output or loss is then formed only from experts in $\mathcal{S}_b$. This differs from fixed ensembling, where every expert contributes to every example with the same global weight. Sample-adaptive routing is especially relevant when inputs have different structure. In multimodal QA, one sample may require OCR and document layout reasoning, another may require chart interpretation and another may depend mainly on natural image recognition. A router can learn to give different experts or teachers different responsibility across these cases. The mathematical distinction can be stated compactly. A global weighting scheme uses one vector $\boldsymbol{\pi} \in \Delta^{N-1}$ for the full training set:

$$\mathcal{L}^{(b)} = \sum_{i=1}^N \pi_i \mathcal{L}_i^{(b)}. \quad (2.28)$$

A sample-adaptive scheme uses $w_i^{(b)}$:

$$\mathcal{L}^{(b)} = \sum_{i=1}^N w_i^{(b)} \mathcal{L}_i^{(b)}, \quad w_i^{(b)} = 0 \text{ for } i \notin \mathcal{S}_b. \quad (2.29)$$

The sparsity constraint limits the number of active experts and makes the teacher-selection decision interpretable. It also reduces the chance that weak or irrelevant teachers dominate the update for a sample.

2.5.5 Routing in Distillation Settings

In distillation, frozen teachers can play the role of experts. The router does not combine hidden activations. Its role is to combine teacher-guided losses. If T_i is teacher i , then teacher T_i contributes a loss $\mathcal{L}_{KD}^{(b,i)}$ for sample b . A routed distillation loss takes the form

$$\mathcal{L}_{route}^{(b)} = \sum_{i \in \mathcal{S}_b} \omega_i^{(b)} \mathcal{L}_{KD}^{(b,i)}. \quad (2.30)$$

This view connects MoE selection to multi-teacher KD: teacher routing is a conditional computation mechanism over supervision signals. Unlike standard MoE, the experts here are frozen, so only the student and router parameters receive gradient updates. Routing in a distillation setting has two additional complications. First, teacher quality is not a fixed property of the teacher alone. It depends on the input, the current student state and the task objective. Second, a high router score does not guarantee that the selected teacher produces a gradient aligned with supervised learning. This means routing is useful but not a complete solution to teacher selection by itself.

2.6 Multi-Teacher Distillation

2.6.1 Weighted Teacher Aggregation

Let $\mathcal{T} = \{T_1, \dots, T_N\}$ be a pool of frozen teachers. For sample b , teacher T_i produces a per-teacher distillation loss $\mathcal{L}_{KD}^{(b,i)}$. In principle, one could average these losses uniformly,

$$\mathcal{L}_{KD}^{(b)} = \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{KD}^{(b,i)}, \quad (2.31)$$

but this assumes every teacher is equally useful for every sample. In practice, different teachers may provide conflicting guidance because their predictions, specialized strengths and KD gradients need not agree on the same training sample. Some teachers may help the supervised objective on a given sample, while others may push the student in a different direction [40]. The sparse-routing framework introduced in Section 2.5 applies directly here by treating teachers as experts whose usefulness varies across samples [20–22]. More generally, an adaptive multi-teacher objective can be written as

$$\mathcal{L}_{KD}^{(b)} = \sum_{i=1}^N w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}, \quad w_i^{(b)} \geq 0, \quad \sum_{i=1}^N w_i^{(b)} = 1. \quad (2.32)$$

Uniform averaging is the special case $w_i^{(b)} = 1/N$. A routed method lets the weights depend on the sample, the student state and the current teacher-selection policy. This is useful when teachers specialize differently: one teacher may be stronger on documents, another on charts and another on natural scene text. The main risk is that adaptive weighting can collapse to one teacher.

2.6.2 Fixed vs. Sample-Dependent Weighting

Existing multi-teacher KD methods differ mainly in how the teacher weights are chosen. A fixed weighting method uses a global vector $\boldsymbol{\pi}$, usually uniform or learned once:

$$w_i^{(b)} = \pi_i, \quad \sum_{i=1}^N \pi_i = 1. \quad (2.33)$$

This is simple and stable, but it cannot express that teacher T_i may be helpful for chart questions and less helpful for document questions. A sample-dependent method predicts $\mathbf{w}^{(b)} = R_\phi(x_b)$, where R_ϕ is a router or teacher-selection policy. Reinforced teacher selection and meta-learned teacher weighting are examples of adaptive multi-teacher KD [20–22].

The distinction matters for heterogeneous VLMs because the teachers may come from different model families. Their strengths can vary by image type, answer format and reasoning demand. A global average can dilute a strong teacher with weaker ones on a sample where specialization matters. A sparse sample-dependent rule keeps only a small teacher set:

$$w_i^{(b)} = \frac{\mathbf{1}\{i \in \mathcal{S}_b\} \exp(r_i^{(b)})}{\sum_{j \in \mathcal{S}_b} \exp(r_j^{(b)})}. \quad (2.34)$$

This keeps aggregation adaptive without dense all-teacher supervision.

2.6.3 Gradient Conflict in Multi-Objective Optimization

Even after sparse routing selects a teacher subset, the remaining teachers may still induce KD gradients that disagree with the supervised objective. To see why, consider the standard combined distillation loss:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{KD}. \quad (2.35)$$

This objective minimizes a supervised term and a teacher-guided term together. Because both terms share the same student parameters, the combined update direction depends on whether the gradients $\nabla_{\theta} \mathcal{L}_{CE}$ and $\nabla_{\theta} \mathcal{L}_{KD}$ agree or conflict. When these two gradient directions are aligned, the combined update reinforces both objectives. When they oppose each other, the resulting update is a compromise that may degrade one or both objectives. This is an instance of *multi-objective optimization*: a single parameter update must serve competing objectives simultaneously [40, 41]. In multi-teacher distillation, the problem is compounded because each teacher contributes its own KD gradient. A routing mechanism can select a small teacher subset, but selection alone cannot guarantee that the remaining KD gradients support the supervised task update.

2.7 Related Work

A rich body of prior work has studied teacher-student transfer, but the existing lines address different parts of heterogeneous VLM distillation. Response-based KD transfers output distributions and has been extended to multimodal compression [11, 12, 30, 42, 43], but it usually assumes a shared tokenizer, a matched output space or a single teacher. Tokenizer-agnostic and optimal-transport methods relax the same-vocabulary assumption by comparing probability mass over different supports [16–18, 31], but they are mainly formulated for single-teacher transfer and do not decide how multiple heterogeneous teachers should be selected or aggregated. Representation matching provides another tokenizer-independent signal through hidden states or similarity measures such as CKA [19], but it is indirect with respect to answer likelihood and does not resolve teacher conflicts. Multi-teacher KD methods show that several teachers can improve student learning [20–22], while sparse expert routing and gradient filtering suggest mechanisms for sample-dependent selection and conflict suppression [35, 36, 39, 40]. However, existing work does not address tokenizer mismatch, sample-specific teacher usefulness and gradient conflict as a single problem for heterogeneous VLM distillation.

Prior work leaves three interrelated gaps. Standard logit-matching objectives break down when teacher and student models use different tokenizers and output vocabularies, while transport-based alternatives that preserve token-string structure remain underexplored for VLMs. Teacher usefulness varies across samples, so global weighting strategies can dilute useful supervision with weaker or conflicting signals. Even a sparse routing mechanism cannot guarantee that selected teachers provide gradients consistent with the supervised task objective. Addressing these gaps together requires treating tokenizer mismatch, sample-adaptive teacher selection and gradient conflict as a single heterogeneous VLM distillation problem.

Chapter 3

Proposed Method

This chapter presents **Gradient-Routed Alignment with Capacity-aware Experts (GRACE)** by first defining the heterogeneous multi-teacher distillation setting, then introducing the **trie-Wasserstein KD loss**, **teacher routing network**, **gradient-agreement refinement mechanism** and **final training objective**. GRACE addresses three core difficulties in heterogeneous teacher-to-student VLM distillation: vocabulary mismatch that makes teacher and student output spaces incomparable, variable teacher usefulness across samples and KD gradients that conflict with the supervised objective. Each component of the method corrects one failure that the previous component cannot fix, so the pieces form a dependency chain, not an independent collection of techniques.

3.1 Method Overview and Problem Formulation

We consider multimodal understanding tasks that require reading image text, understanding the question and generating an answer sequence [4–6]. At its core, GRACE separates two questions that most KD methods conflate: what does the teacher believe and should the student trust this teacher here. Trie-Wasserstein KD makes teacher beliefs comparable across tokenizers, sparse routing asks which teachers are relevant for this sample and gradient-agreement refinement asks which of those teachers is currently compatible with the supervised objective. The training set is

$$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N_{\text{data}}}, \quad (3.1)$$

where x_n is a multimodal input and y_n is the target answer sequence. Let S_θ denote the student VLM with trainable parameters θ , let G_ϕ denote the trainable teacher router with parameters ϕ and let $\mathcal{T} = \{T_i\}_{i=1}^N$ be the frozen teacher pool. For each sample b , the student produces logits $\mathbf{z}_s^{(b)}$ and teacher T_i provides teacher logits $\mathbf{z}_t^{(b,i)}$.

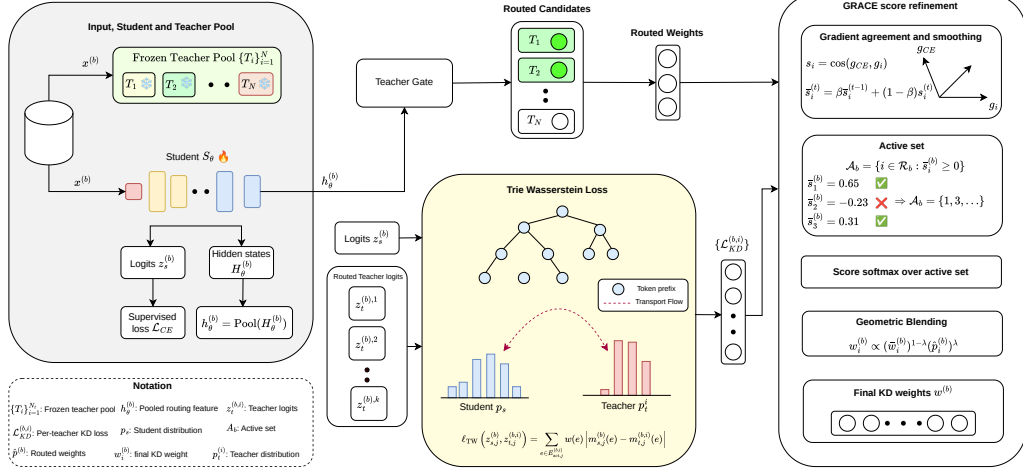


Figure 3.1: Overview of the GRACE architecture.

Throughout this chapter, the frozen teachers are treated as tokenizer-specific experts: each teacher may use a different tokenizer, vocabulary and model family and the router selects which experts contribute to each training sample. The full training pipeline is summarized in Figure 3.1. The method is organized around four components. **Trie-Wasserstein KD** computes per-teacher distillation losses over a shared byte trie, enabling tokenizer-agnostic output matching. The **teacher routing network** selects a sparse subset of candidate teachers from student-side hidden states at each training step. **Gradient-agreement refinement** adjusts the routed teacher weights based on how well each teacher’s distillation gradient aligns with the supervised objective. Finally, the **training objective** combines supervised cross-entropy, routed KD and router auxiliary losses into a single end-to-end update. These components are ordered by dependency: trie-Wasserstein makes cross-tokenizer comparison possible but does not choose which teacher to use. Routing chooses teachers but does not check whether their gradients help. Gradient-agreement refinement corrects routed weights but still requires trie-Wasserstein to produce meaningful per-teacher losses in the first place. Together, they extend sparse expert routing and token-level logit supervision to the heterogeneous multi-teacher VLM setting, where teachers and student may use different tokenizers and vocabularies [35, 36, 40]. The main notation used throughout the method is listed in Table 3.1. Section 3.1.1 walks through one complete update before the remaining sections define each component formally.

Table 3.1: Main notation.

Symbol	Meaning
$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N_{\text{data}}}$	Multimodal samples and target answer sequences.
$S_\theta, \mathcal{T} = \{T_i\}_{i=1}^N$	Trainable student VLM and frozen teacher pool.
G_ϕ	Trainable teacher routing network with parameters ϕ .
T_{train}	Total number of training steps.
$\mathbf{z}_s^{(b)}, \mathbf{z}_t^{(b,i)}$	Student and teacher logits for sample b .
$\mathbf{H}_s, \mathbf{h}$	Student hidden states and pooled routing feature.
$\rho_{\text{trie}}, K_{\text{TW}}, e_{\text{tail}}$	Trie edge decay, sparse support size and residual tail edge.
k_{route}	Configured router top- k .
$\mathbf{r}, \mathbf{p}$	Router logits and soft router weights.
$\mathcal{C}_b, \mathcal{R}_b, \hat{\mathbf{p}}^{(b)}$	Top- k candidate teacher set, routed teacher set and renormalized routed weights.
$s_i, \hat{s}_i, \tilde{a}_i, a_i^{(b)}, \mathcal{A}_b$	Raw agreement score, EMA-smoothed agreement score, teacher active flag, sample active flag and active routed set.
$\omega^{(b)}, \mathbf{w}^{(b)}$	Score-softmax weights and final KD weights.
$\Omega_b^{CE}, \Omega_{b,i}^{KD}$	Supervised CE positions and matched KD positions.
$\mathbf{g}_{CE}, \mathbf{g}_i$	CE and teacher-KD parameter gradients used by GRACE.
$\beta_{\text{EMA}}, \beta_{\text{score}}, \epsilon_{\text{num}}, \epsilon_{\text{fb}}, \lambda_{\text{blend}}$	EMA decay, score-softmax sharpness, cosine stabilizer, fallback threshold and router-score blend exponent.
TopKRoute	Sparse top- k teacher routing with routed-weight renormalization.
Uniform fallback	Uniform weights over the active routed set.
Blend	Mixing routed weights with GRACE score weights.
EMA	Exponential moving average for GRACE scores.

3.1.1 Training Flow

A GRACE update starts from a mini-batch $\mathcal{B} = \{(x_b, y_b)\}_{b=1}^B$ and a frozen teacher pool $\mathcal{T} = \{T_i\}_{i=1}^N$. The student first produces answer-token logits, hidden states and the supervised cross-entropy loss. The hidden states serve a dual role: they participate in the CE computation and act as a compact description of the sample that drives teacher routing. Supervised answer positions define both the CE objective and the hidden-state region pooled by the router. Sparse routing selects a teacher subset $\mathcal{R}_b$, trie-Wasserstein KD computes per-teacher losses $\mathcal{L}_{KD}^{(b,i)}$ and gradient-agreement refinement converts routing scores into final KD weights. The router’s answer to which teachers are relevant and the agreement check’s answer to which teachers are compatible together ensure that only useful teacher signals reach the optimization. The update ends by combining supervised CE, routed KD and router regularization terms into one optimization objective.

3.2 Trie-Wasserstein Cross-Tokenizer Distillation

3.2.1 Supervised Answer Tokens

Distillation is applied only on **supervised answer tokens**, not on prompt tokens. For sample b , let Ω_b^{CE} denote the student positions whose labels are not ignored. For teacher T_i , let $\Omega_{b,i}^{KD}$ denote the matched answer positions after masking ignored labels on both student and teacher sides and truncating both spans to a common length. This is a position-wise approximation across tokenizers: after answer-span preprocessing, the j -th student and teacher answer positions are treated as corresponding positions. Trie-Wasserstein matching handles the token-distribution mismatch at each aligned position, while sequence-level byte alignment remains outside the objective. Restricting distillation to answer positions avoids wasting capacity on prompt formatting, padding and instruction wording, where teacher and student distributions carry no task-relevant signal.

3.2.2 Shared Byte Trie

For each sample b , token position $j \in \Omega_{b,i}^{KD}$ and teacher T_i , let $\mathbf{z}_{s,j}^{(b)} \in \mathbb{R}^{V_s}$ be the student logits and $\mathbf{z}_{t,j}^{(b,i)} \in \mathbb{R}^{V_i}$ be the teacher logits. The student and teacher probability distributions are

$$\mathbf{p}_{s,j}^{(b)} = \text{softmax}\left(\frac{\mathbf{z}_{s,j}^{(b)}}{\tau_s}\right), \quad \mathbf{p}_{t,j}^{(b,i)} = \text{softmax}\left(\frac{\mathbf{z}_{t,j}^{(b,i)}}{\tau_t}\right), \quad (3.2)$$

where τ_s and τ_t are the student and teacher temperatures.

Direct token-index matching assumes that bin 17 in the teacher vocabulary means the same thing as bin 17 in the student vocabulary. That assumption fails entirely when tokenizers differ. GRACE maps decoded teacher and student token strings into a **shared UTF-8 byte trie**, so tokens live in a shared prefix tree, not arbitrary vocabulary-index space. Tokens with common byte prefixes share the same initial trie path even if their tokenizer IDs are unrelated. UTF-8 bytes provide a fixed alphabet and expose tokenizer artifacts such as leading spaces, wordpiece prefixes and multimodal special tokens. Character-level matching is less stable because the same visible text may have different Unicode code-point sequences. Concretely, a token with a leading space maps to bytes [32, 99, 97, 116], while `cat` maps to [99, 97, 116]: the same visible string can have a different byte sequence, so byte-level comparison is necessary for consistent matching.

Each token string is converted to a byte sequence and terminated with an EOS sentinel. The sentinel separates complete tokens from strict prefixes, so `doc` and `document` remain distinct leaves even though they share an initial path. The practical effect is a shift from matching identical token IDs to keeping probability mass on similar decoded strings close, giving all tokenizers a common comparison space. The trie is therefore more than a tokenizer-compatibility device: it defines a geometry over decoded token strings, illustrated in Figure 3.2 for the pairs `document/receipt` and `doc/rece`. Tokens such as `doc`, `document` and `documentation` are close because they share deep trie paths, while tokens such as `chart`, `invoice` and `red` diverge immediately. That geometry is well suited to OCR-heavy VQA, where correct and near-correct answers often differ through prefixes, suffixes, spaces or number formats while remaining lexically nearby. For a normal byte-trie or EOS edge e , let $d(e)$ be its depth. GRACE assigns depth-based edge weights as follows:

$$w(e) = \rho_{\text{trie}}^{d(e)-1}, \quad (3.3)$$

where $0 < \rho_{\text{trie}} < 1$. The first edge from the root has weight 1 and deeper edges decay geometrically. This makes shared prefixes cheaper to transport across than unrelated byte paths: two tokens that agree on the first three bytes pay much less than two tokens with no overlap at all. The loss therefore penalizes how far the student’s probability mass must move through the shared lexical tree to resemble the teacher’s mass, not whether the student assigns probability to the same token indices. The decay still preserves a valid tree metric [17, 32] and different tokenizers may stop at different depths while still sharing the same byte-prefix edges whenever their decoded token strings overlap in byte-prefix structure.

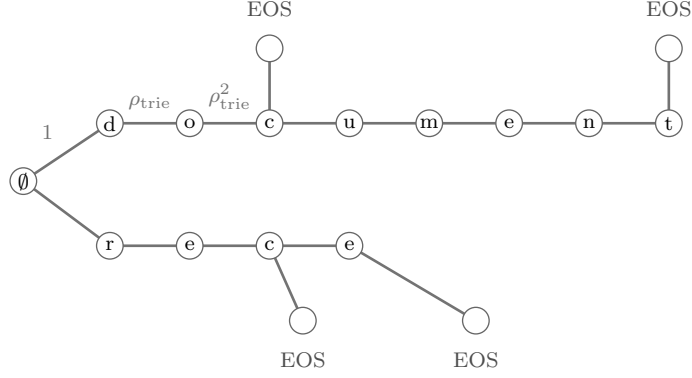


Figure 3.2: Shared-trie tokenizer matching.

In the shared-trie example of Figure 3.2, shorter teacher token sequences terminate earlier at EOS, whereas longer student token sequences extend farther down the trie. Tokenizer mismatch is therefore represented through shared prefix structure, not direct token-ID comparison, preserving lexical evidence that would be lost in vocabulary-index space.

3.2.3 Trie-Wasserstein Objective

The shared byte trie induces a tree metric over decoded token strings. Let $\mathcal{V}_s$ and $\mathcal{V}_t$ denote the student and teacher token vocabularies being compared at a given answer position and let $\pi(v)$ denote the trie path of decoded token v . For an edge e , the student and teacher subtree masses are

$$m_s(e) = \sum_{\substack{v \in \mathcal{V}_s \\ e \in \pi(v)}} p_s(v), \quad m_t(e) = \sum_{\substack{u \in \mathcal{V}_t \\ e \in \pi(u)}} p_t(u).$$

$m_s(e)$ and $m_t(e)$ measure how much probability mass each distribution assigns to token paths passing through edge e . The W_1 optimal transport distance between teacher and student token distributions on the trie can then be written as a weighted sum of subtree-mass imbalances. If teacher and student both concentrate mass on strings beginning with `doc`, they are already aligned along that branch. If one puts mass on `doc` and the other on `abc`, the mass diverges near the root and the transport cost is high. The loss rewards lexical closeness, not exact token-ID equality:

$$W_T(p_s, p_t) = \sum_{e \in E} w(e) |m_s(e) - m_t(e)|. \quad (3.4)$$

This form follows from the standard W_1 Kantorovich optimal transport problem under the trie-induced tree metric [17, 32]. The sparse construction below applies the same formula to a bounded top- K_{TW} support with a residual tail edge.

3.2.4 Sparse Support and Residual Tail Edge

Equation (3.4) is exact on the complete shared trie, but evaluating every vocabulary token at every answer position is expensive. GRACE builds a **sparse augmented trie**, so most useful distributional information, concentrated in high-probability tokens, is captured exactly. The tail edge handles the remainder: without it, omitted mass would silently disappear, making the teacher appear more confident than it really is and distorting the transport distance. At each aligned answer position, it masks ignored vocabulary entries, retains the **top- K_{TW} student and teacher tokens** and inserts those tokens with their exact byte paths. Probability mass outside the retained sets is not discarded. It is collapsed into a single **residual tail edge**. This gives an exact distance on the collapsed sparse trie and it approximates the complete trie distance by representing all omitted vocabulary mass with one tail atom.

Let K_{TW} denote the sparse trie support size. Let $\mathcal{V}_s^*$ and $\mathcal{V}_t^*$ denote the non-ignored student and teacher vocabularies. Before constructing the sparse trie, the student and teacher probabilities are renormalized over the non-ignored vocabularies so that the retained and residual masses form valid distributions. Let $\mathcal{K}_{s,j}^{(b)} \subset \mathcal{V}_s^*$ and $\mathcal{K}_{t,j}^{(b,i)} \subset \mathcal{V}_t^*$ be the retained token sets. The residual mass assigned to the tail edge is

$$r_{s,j}^{(b)} = 1 - \sum_{v \in \mathcal{K}_{s,j}^{(b)}} p_{s,j}^{(b)}(v), \quad r_{t,j}^{(b,i)} = 1 - \sum_{v \in \mathcal{K}_{t,j}^{(b,i)}} p_{t,j}^{(b,i)}(v). \quad (3.5)$$

Let $\pi(v)$ be the trie path for token v and let e_{tail} be the global residual tail edge with $w(e_{\text{tail}}) = 1$. For any active trie edge e , the sparse subtree masses are

$$m_{s,j}^{(b)}(e) = \sum_{\substack{v \in \mathcal{K}_{s,j}^{(b)} \\ e \in \pi(v)}} p_{s,j}^{(b)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{s,j}^{(b)}, \quad (3.6)$$

$$m_{t,j}^{(b,i)}(e) = \sum_{\substack{v \in \mathcal{K}_{t,j}^{(b,i)} \\ e \in \pi(v)}} p_{t,j}^{(b,i)}(v) + \mathbf{1}[e = e_{\text{tail}}] r_{t,j}^{(b,i)}. \quad (3.7)$$

The token-level trie-Wasserstein loss is

$$\ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}) = \sum_{e \in \mathcal{E}_{act,j}^{(b,i)}} w(e) \left| m_{s,j}^{(b)}(e) - m_{t,j}^{(b,i)}(e) \right|, \quad (3.8)$$

where $\mathcal{E}_{act,j}^{(b,i)}$ contains the active student edges, active teacher edges and the residual tail edge. Before the absolute value is applied, $m_{s,j}^{(b)}(e) - m_{t,j}^{(b,i)}(e)$ is positive when edge e carries more student mass and negative when it carries more teacher mass.

The per-sample, per-teacher KD loss is the mean over supervised answer tokens:

$$\mathcal{L}_{KD}^{(b,i)} = \frac{1}{|\Omega_{b,i}^{KD}|} \sum_{j \in \Omega_{b,i}^{KD}} \ell_{TW}(\mathbf{z}_{s,j}^{(b)}, \mathbf{z}_{t,j}^{(b,i)}) . \quad (3.9)$$

Because the tail edge accounts for all omitted vocabulary mass, the sparse trie loss is exact on the collapsed structure without simply dropping the tail.

Complexity analysis. The shared byte trie is built once with time and memory cost $O(V_s \bar{L}_s + V_t \bar{L}_t)$, where V_s and V_t are the student and teacher vocabulary sizes and $\bar{L}_s$ and $\bar{L}_t$ are the average byte-path lengths. For one aligned answer position, extracting the top- K_{TW} student and teacher tokens costs $O(V_s \log K_{TW} + V_t \log K_{TW})$ with heap-based selection. Given the retained token sets, sparse subtree-mass accumulation walks only the retained token paths and the residual tail edge, costing $O(K_{TW} \bar{L}_s + K_{TW} \bar{L}_t + A)$, where A is the number of active trie edges. If the implementation explicitly sorts or deduplicates active edge indices, this adds an $O(A \log A)$ term. Otherwise, indexed accumulation gives an $O(A)$ edge-aggregation cost. Runtime memory is $O(E + P_s + P_t + A)$, where E is the number of trie edges and P_s and P_t are the stored student and teacher path-index arrays. Unlike dense optimal transport, the sparse computation does not materialize a $|V_s| \times |V_t|$ transport matrix. This gives a tokenizer-agnostic loss that compares probability mass on shared trie edges and avoids direct token-index matching. The top- K_{TW} +tail approximation keeps this structured transport view bounded by the retained support.

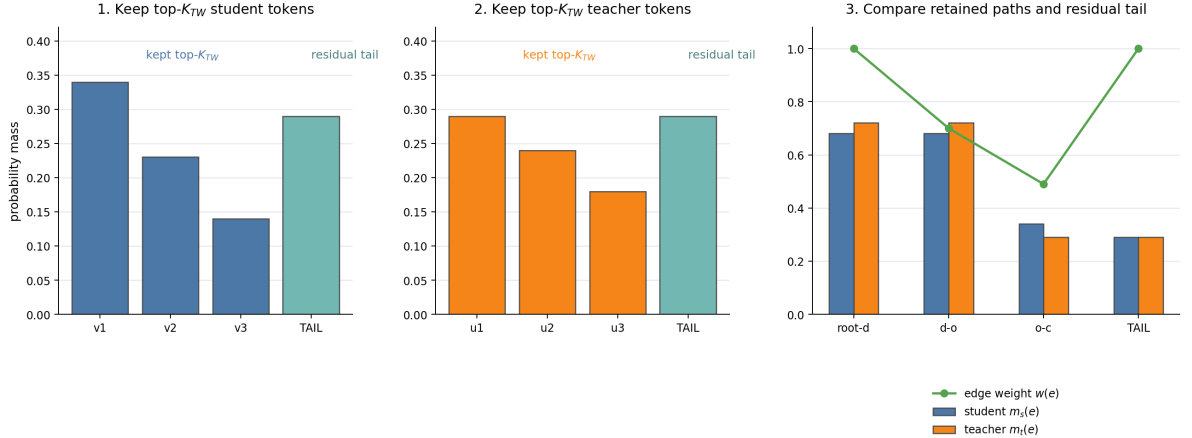


Figure 3.3: Sparse trie-Wasserstein support.

Figure 3.3 shows the retained student and teacher supports, the residual tail edge and the signed subtree-mass comparison. Prefix sharing makes pairs such as `doc/document` close in the trie, while unrelated strings such as `abc/def` diverge immediately.

3.3 Teacher Routing Network

3.3.1 Router Architecture

With multiple teachers, uniform averaging ignores sample-specific teacher strengths: a document question, a chart question and a scene-text question may each benefit from a different expert and the student’s own hidden states are the most direct signal of which kind of example is being processed. This makes GRACE routing fundamentally different from sparse mixture-of-experts routing in standard transformers, where routing is a computation decision: which expert runs the forward pass. In GRACE, the experts are frozen external models and routing is a supervision decision: which teacher shapes the student’s learning signal. GRACE therefore attaches a **teacher routing network** to the linear language-model head, so the gate operates on the text hidden state immediately before token prediction. The full pipeline from pooled student features to sparse top- k teacher selection is shown in Figure 3.4.

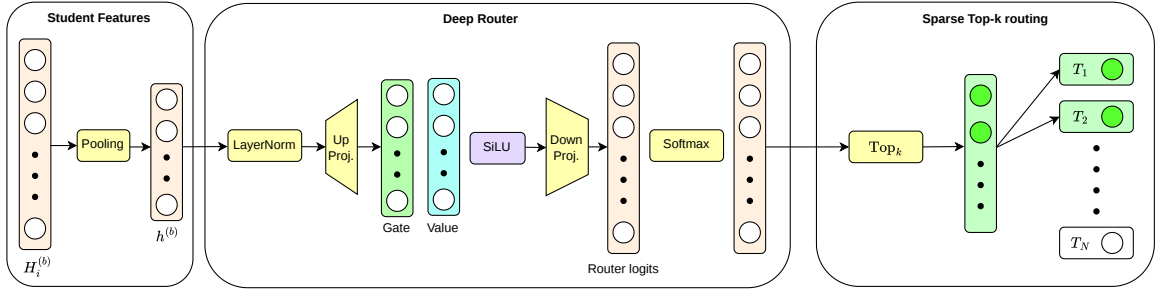


Figure 3.4: Teacher routing network.

Let $\mathbf{H}_s^{(b)}$ denote the student hidden states captured immediately before the language-model head for sample b . These are pooled over supervised answer positions to form the **routing feature**:

$$\mathbf{h}^{(b)} = \text{Pool}(\mathbf{H}_s^{(b)}) . \quad (3.10)$$

Pooling over answer positions captures what the student currently represents about the output, which is more informative than pooling over the full sequence. The routing feature is also student-state-dependent, not only dataset-dependent: the same image-question pair can produce a different routing decision as the student’s representations evolve during training, allowing teacher selection to adapt continuously. The operation uses the mean over supervised answer positions and requires at least one supervised answer token per sample. A learned SwiGLU-style MLP router $G_\phi(\cdot)$ maps the routing feature to gate logits,

$$\mathbf{r}^{(b)} = G_\phi(\mathbf{h}^{(b)}) . \quad (3.11)$$

which are converted to soft routing weights by softmax:

$$\mathbf{p}^{(b)} = \text{softmax}(\mathbf{r}^{(b)}), \quad (3.12)$$

where $p_i^{(b)}$ is the initial gate weight for teacher T_i . The router then masks to the top- k_{route} teachers and renormalizes the selected weights. The internal stack uses LayerNorm, an up-projection, a gate-value split, SiLU gating and a down-projection, keeping score computation dense while teacher use remains sparse after top- k masking. For a mini-batch, the captured hidden state has shape $B \times L \times d_h$ and masked mean pooling returns a $B \times d_h$ routing feature. All reported experiments use $N = 4$ teachers, so the router hidden width is set to $d_r = 64$. After LayerNorm, the up-projection maps $\mathbb{R}^{d_h}$ to $\mathbb{R}^{2d_r}$, producing a tensor that is split into gate and value components in $\mathbb{R}^{d_r}$. Applying $\text{SiLU}(\text{gate}) \odot \text{value}$ and a down-projection from $\mathbb{R}^{d_r}$ to $\mathbb{R}^N$ then yields one logit per teacher. Wider hidden sizes in the codebase are scaling fallbacks for larger teacher pools and play no role in the reported experiments, whose cost is dominated by four-teacher logit storage and routed KD computation.

3.3.2 Top-k Routing

Soft routing weights are dense, but distillation should not use every teacher for every sample. GRACE converts them into a **sparse routed set** by selecting the highest-scoring teachers for sample b :

$$\mathcal{C}_b = \text{TopK}(\mathbf{r}^{(b)}, \min(k_{\text{route}}, N)). \quad (3.13)$$

The candidate set $\mathcal{C}_b$ serves as the routed teacher set $\mathcal{R}_b$. With $m_i^{(b)} = \mathbf{1}[i \in \mathcal{C}_b]$ denoting the top- k assignment mask, non-selected weights are zeroed and the remaining softmax weights are renormalized:

$$\hat{p}_i^{(b)} = \frac{m_i^{(b)} p_i^{(b)}}{\sum_{j=1}^N m_j^{(b)} p_j^{(b)}}. \quad (3.14)$$

Dense averaging can dilute a strong teacher with weak or irrelevant ones. Top- k routing makes supervision sparse and sample-specialized. Routing alone is not sufficient, however. A teacher can look relevant because its domain matches the input while its gradient still conflicts with the supervised target. Figure 3.5 contrasts dense gate probabilities with the sparse routed assignment. The logged assignment rate records how often each teacher appears in the routed set. For example, a document-specialized teacher may assign high probability to a plausible but incorrect field value. Gradient-agreement refinement, introduced in Section 3.4, addresses this residual failure. This sparse routed teacher set keeps the per-sample distillation cost bounded by k_{route} teachers and provides the candidate pool that refinement operates on.

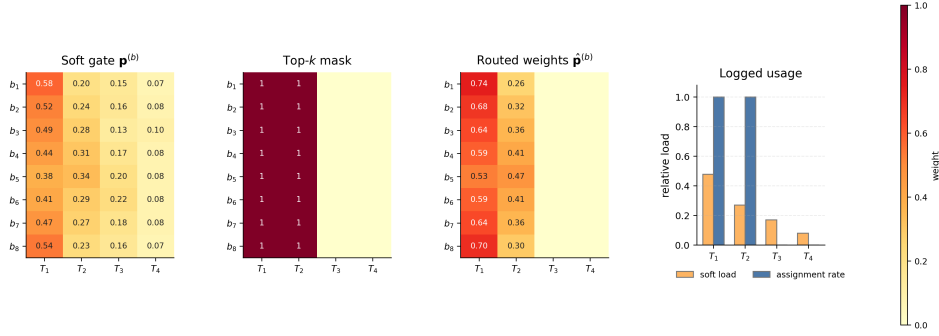


Figure 3.5: Top- k routing dynamics.

3.3.3 Router Entropy and Z-Loss

Two auxiliary terms regularize teacher selection. **Entropy regularization** is implemented as a negative-entropy term that discourages premature routing collapse:

$$\mathcal{L}_{ent} = -\frac{1}{B} \sum_{b=1}^B H(\mathbf{p}^{(b)}). \quad (3.15)$$

The **router z-loss** keeps router logits numerically well behaved:

$$\mathcal{L}_z = \frac{1}{B} \sum_{b=1}^B \left(\log \sum_{i=1}^N e^{r_i^{(b)}} \right)^2. \quad (3.16)$$

Both terms are pure stabilizers: entropy discourages the router from collapsing onto a single teacher too early, while z-loss keeps routing logits numerically controlled. Neither competes with the CE or KD objectives.

3.4 Gradient-Agreement Refinement

Routed teachers may still conflict with the supervised objective after teacher selection. **Gradient-agreement refinement** scores each teacher’s KD direction against the supervised CE direction, then uses those scores only inside the routed candidate set. Routing answers which teacher seems relevant to this sample. Gradient agreement answers which of those teachers is currently compatible with learning the correct answer, acting as the method’s truth check on the router’s nominations. The two checks are complementary because relevance does not guarantee gradient compatibility.

3.4.1 Gradient Agreement Scoring

Let $\Theta \subseteq \theta$ denote the trainable student parameters used by the optimizer. To score each teacher, GRACE compares its **teacher-specific KD gradient** against the **supervised CE gradient** in student-parameter space. The CE gradient is

$$\mathbf{g}_{CE} = \nabla_{\Theta} \mathcal{L}_{CE}. \quad (3.17)$$

For teacher T_i , the KD gradient comes from that teacher’s mean distillation loss over the full mini-batch:

$$\mathbf{g}_i = \nabla_{\Theta} \left(\frac{1}{B} \sum_{b=1}^B \mathcal{L}_{KD}^{(b,i)} \right). \quad (3.18)$$

Here, $\mathcal{L}_{KD}^{(b,i)}$ is the trie-Wasserstein loss from Equation 3.9. These gradients only score teacher compatibility. The final parameter update is still computed from the full training objective. A teacher whose KD gradient points opposite to the CE gradient is pulling the student away from the correct answer, even if the router selected it confidently. The first-order parameter update makes this explicit: $\Delta\theta \propto -(\mathbf{g}_{CE} + \alpha \sum_i w_i \mathbf{g}_i)$. When $\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle > 0$, teacher T_i reinforces the supervised direction. When $\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle < 0$, its contribution partially opposes the supervised update and increases the first-order CE loss when the goal is to reduce it. Cosine similarity therefore approximates whether this teacher’s KD update will help or hurt the supervised objective in the next step.

3.4.2 Agreement Scores and Active Set

Directional agreement between supervised learning and KD is measured with cosine similarity:

$$s_i = \frac{\langle \mathbf{g}_{CE}, \mathbf{g}_i \rangle}{\max(\|\mathbf{g}_{CE}\|_2 \|\mathbf{g}_i\|_2, \epsilon_{\text{num}})}. \quad (3.19)$$

The agreement score s_i is **teacher-level** within the current mini-batch and is broadcast across samples. Sample dependence enters through the routed set $\mathcal{R}_b$, routed weights $\hat{\mathbf{p}}^{(b)}$ and active set $\mathcal{A}_b$. To combine teacher-level agreement with sample-specific routing, GRACE first converts the agreement score into a binary teacher flag:

$$\tilde{a}_i = \mathbf{1}[s_i > \delta], \quad (3.20)$$

where δ is the GRACE threshold. This flag is then intersected with the routed teacher set for each sample:

$$a_i^{(b)} = \mathbf{1}[i \in \mathcal{R}_b] \tilde{a}_i, \quad \mathcal{A}_b = \{i \in \mathcal{R}_b : a_i^{(b)} = 1\}. \quad (3.21)$$

The intersection with $\mathcal{R}_b$ keeps the search inside the routed candidates: routing narrows the pool first and gradient agreement filters within it. Computing agreement at the teacher level over the full mini-batch, not at the sample level, is a deliberate trade-off. A sample-level agreement score would be more precise but prohibitively noisy and expensive. The batch-level score is cheaper and more stable, while sample-level adaptation is preserved through the routed set $\mathcal{R}_b$. The final active set $\mathcal{A}_b$ intersects both signals: batch-level objective compatibility from gradient agreement and sample-level relevance from routing. The threshold δ is applied to the raw score s_i , while the EMA-smoothed score $\hat{s}_i$ controls score-softmax weighting and fallback spread. Agreement scores act as non-differentiable routing statistics, so the final optimization step does not backpropagate through the gradient-agreement computation itself or treat the scoring procedure as an additional trainable path.

3.4.3 Warmup, EMA Smoothing and Score Weights

Teacher weights can change abruptly when agreement scores are estimated from a single mini-batch. GRACE stabilizes this step with **warmup**, **EMA smoothing** and **score-softmax weighting**. Let $\rho_g \in [0, 1]$ denote the warmup ratio relative to the total number of training steps. During the first $\rho_g T_{\text{train}}$ steps, routed gate weights are left unchanged:

$$w_i^{(b)} = \hat{p}_i^{(b)}. \quad (3.22)$$

Early in training, the student’s representations and the router’s assignments are both unstable, so gradient-based filtering applied too soon risks rejecting useful teachers because the agreement scores themselves are meaningless. Warmup prevents the agreement mechanism from overfitting to early gradient noise and missing genuine teacher usefulness. After warmup, routed teachers are refined with a smoothed score distribution. Let $\bar{s}_i^{\text{prev}}$ be the running exponential moving average (EMA) of past batch means for teacher T_i and let β_{EMA} be its decay. The smoothed score is

$$\hat{s}_i = \beta_{\text{EMA}} \bar{s}_i^{\text{prev}} + (1 - \beta_{\text{EMA}}) s_i. \quad (3.23)$$

EMA smoothing turns gradient agreement from a momentary vote into a reputation score. A teacher’s raw score on one batch answers whether it was helpful this step, while the smoothed score answers whether it has been consistently helpful recently. This reduces oscillation in teacher weights and makes the scoring more robust to difficult batches on which even a useful teacher may temporarily misalign. For nonempty active sets, smoothed scores are converted into a score-softmax weight vector over $\mathcal{A}_b$:

$$\omega_i^{(b)} = \frac{\exp(\beta_{\text{score}} \hat{s}_i) \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N \exp(\beta_{\text{score}} \hat{s}_j) \mathbf{1}[j \in \mathcal{A}_b]}, \quad (3.24)$$

β_{score} controls how sharply the highest-scoring active teacher is preferred, setting up the fallback and blending cases below.

3.4.4 Weight Blending and Fallback

Final KD weights are assigned through three mutually exclusive cases for each sample. The three cases implement a calibrated decision rule. When evidence is missing, GRACE uses the router fallback. When evidence is weak, it applies uniform active weights and does not treat a small score difference as meaningful. When evidence is confident, agreement contributes directly to the final weights. This calibration avoids overcommitting to weak gradient signals while still allowing strong agreement evidence to shape teacher weighting.

No active teacher. If $\mathcal{A}_b = \emptyset$, the routed weights are retained:

$$w_i^{(b)} = \hat{p}_i^{(b)}. \quad (3.25)$$

Low-confidence scores. If active teachers exist but their smoothed scores are too close, uniform weights are used over the active set. The score range is

$$\Delta^{(b)} = \max_{i \in \mathcal{A}_b} \hat{s}_i - \min_{i \in \mathcal{A}_b} \hat{s}_i. \quad (3.26)$$

If $\mathcal{A}_b \neq \emptyset$ and $\Delta^{(b)} < \epsilon_{\text{fb}}$, then

$$w_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{A}_b]}{|\mathcal{A}_b|}. \quad (3.27)$$

Confident scores. Otherwise, router weights are blended with score weights. Let $\lambda_{\text{blend}} \in [0, 1]$ be the router-blend exponent. The unnormalized blended weights are

$$\tilde{w}_i^{(b)} \propto \left(\hat{p}_i^{(b)}\right)^{\lambda_{\text{blend}}} \left(\omega_i^{(b)}\right)^{1-\lambda_{\text{blend}}}, \quad (3.28)$$

followed by normalization over the active routed teachers:

$$w_i^{(b)} = \frac{\tilde{w}_i^{(b)} \mathbf{1}[i \in \mathcal{A}_b]}{\sum_{j=1}^N \tilde{w}_j^{(b)} \mathbf{1}[j \in \mathcal{A}_b]}. \quad (3.29)$$

Equations 3.24, 3.27 and 3.29 define the gradient-refinement operator: the router chooses $\mathcal{R}_b$, agreement filtering narrows it to $\mathcal{A}_b$ and final weights are retained from the router, assigned uniformly or geometrically blended depending on confidence. Neither signal is trusted alone because they measure different things: router weights capture sample-specific relevance, while agreement weights capture objective compatibility. A teacher could be globally gradient-aligned but irrelevant to a specific sample type, while another could be sample-relevant but currently conflicting with the supervised objective. The geometric blend respects both, with λ_{blend} controlling how much the router’s domain judgment is retained against the agreement signal. These regimes are shown in Figure 3.6, where the three cases separate router fallback, uniform active weighting and confident score-based blending.



Figure 3.6: Representative KD weighting regimes.

This design keeps teacher selection sparse throughout: gradient-agreement refinement can narrow the routed set but never expands it, so KD cost remains bounded by k_{route} teachers per sample.

3.5 Routed KD Aggregation and Final Objective

The final teacher weights $w_i^{(b)}$ combine the routed teacher set $\mathcal{R}_b$, router weights $\hat{\mathbf{p}}^{(b)}$ and gradient-agreement scores. The objective aggregates routed per-teacher KD losses into a sample-level term, averages those terms across the mini-batch and then adds supervised and router-regularization losses.

Sample-level aggregation. For sample b , only teachers in the routed set $\mathcal{R}_b$ contribute to the KD loss:

$$\bar{\mathcal{L}}_{KD}^{(b)} = \sum_{i \in \mathcal{R}_b} w_i^{(b)} \mathcal{L}_{KD}^{(b,i)}. \quad (3.30)$$

Non-routed teachers have zero contribution, while routed teachers are weighted by the fallback, uniform-active or blended rule. All post-routing refinement decisions enter the optimization through $w_i^{(b)}$.

Batch-level aggregation. The mini-batch KD term averages the sample-level KD losses:

$$\mathcal{L}_{KD} = \frac{1}{B} \sum_{b=1}^B \bar{\mathcal{L}}_{KD}^{(b)}. \quad (3.31)$$

This scalar term supplies teacher supervision to the final objective. It stays compatible with heterogeneous teachers because each per-teacher loss uses trie-Wasserstein matching, which is tokenizer-agnostic, with no direct token-index KL divergence. The role of each term in Equation (3.32) follows from the dependency chain introduced in Section 3.1: $\mathcal{L}_{CE}$ is the compass that fixes the learning direction, $\mathcal{L}_{KD}$ provides the teacher map, routing chooses which map to consult and gradient agreement checks whether the map currently points in the compass direction.

Final objective. The complete GRACE objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{KD}\mathcal{L}_{KD} + \lambda_{ent}\mathcal{L}_{ent} + \lambda_z\mathcal{L}_z, \quad (3.32)$$

In this objective, $\mathcal{L}_{CE}$ anchors the student to supervised answer tokens, $\mathcal{L}_{KD}$ transfers routed teacher supervision and $\mathcal{L}_{ent}$ together with $\mathcal{L}_z$ regularize the teacher routing network. The coefficient α_{KD} controls the strength of teacher supervision, while λ_{ent} and λ_z control the auxiliary router losses. Only the student parameters θ and router parameters ϕ are optimized, while all teacher parameters remain frozen. Together, the method can be read as a three-layer defense: trie-Wasserstein defends against tokenizer mismatch, sparse routing defends against irrelevant teachers and gradient-agreement refinement defends against selected teachers whose gradients conflict with supervised learning. The result is a training update that is tokenizer-aware, sample-specialized and objective-compatible.

Chapter 4

Evaluation Methodology

This chapter describes the experimental setup used to evaluate GRACE and all compared methods. It covers the student models, benchmarks, training configuration, evaluation metrics and baselines. The ablation settings and implementation details needed to reproduce the main results are included at the end of the chapter.

4.1 Experimental Setup

All experiments follow a shared evaluation protocol for fair comparison across methods. The experiments instantiate the student side with **SmolVLM-500M** and **SmolVLM-256M**, evaluated on multimodal benchmarks [15]. All models are evaluated with **VLM-Eval-Kit** [44], which provides a shared inference, post-processing and metric-computation pipeline for the three benchmarks. VLM-Eval-Kit uses a generation-based evaluation interface. For each benchmark sample, the toolkit loads the image-question pair, applies the dataset-specific prompt construction logic and calls the model’s generation or chat interface to produce a free-form text answer. This choice is important for VLMs because it evaluates the same interface used at inference time and avoids replacing answer generation with likelihood scoring over a fixed set of options. After generation, the toolkit writes model predictions to result files and invokes the benchmark-specific evaluator.

The post-processing stage depends on the task format. For structured yes/no or multiple-choice tasks, VLM-Eval-Kit can use exact matching or an LLM-based choice extractor to recover the final option from the generated text. For open-answer VQA benchmarks such as the ones used here, the generated answer is normalized before scoring so that superficial formatting differences do not dominate the metric. The final score is then computed with the official metric associated with each dataset. For example, a TextVQA prediction is matched against the human answer set using VQA accuracy, so a short answer such as **stop** is credited according to annotator agreement. A DocVQA prediction is scored with ANLS, so a near miss such as **INV-4572** versus **4572** receives partial credit only when the normalized edit similarity is high enough. A ChartQA numeric answer is evaluated with relaxed accuracy, so a value within 5% of the reference number is counted as correct. In this thesis, using the same toolkit for all methods fixes the prompt template, decoding interface, answer normalization and metric implementation, so evaluation-code differences do not drive the reported performance gaps.

4.2 Datasets

We evaluate three text-centric VQA benchmarks: **TextVQA**, **DocVQA** and **ChartQA**. These benchmarks test whether the model can read visual text, understand questions and perform basic reasoning across natural images, documents and charts. They were chosen because they cover complementary visual domains and combine OCR-heavy understanding with layout interpretation and numerical reasoning [4–6].

TextVQA tests question answering on natural images containing scene text, requiring the model to read and reason over text embedded in real-world scenes [4]. *Sample question*: “What word is written on the stop sign?”

DocVQA tests question answering on scanned document images, where layout, paragraphs, tables and forms are central to answering correctly [5]. *Sample question*: “What is the invoice number shown in this document?”

ChartQA tests question answering over charts, combining chart reading with numerical and logical reasoning [6]. *Sample question*: “Which year has the highest sales value in the chart?”

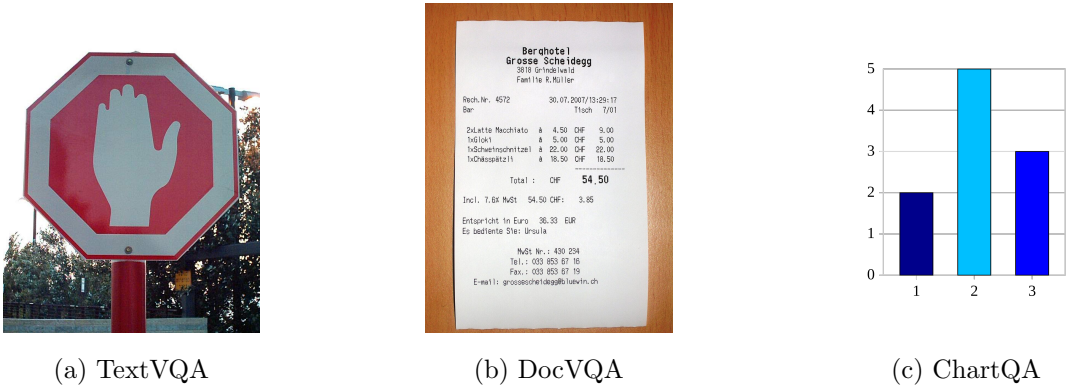


Figure 4.1: Representative dataset examples.¹

Figure 4.1 shows the visual differences behind this benchmark choice. The TextVQA panel represents natural-image reading, where the relevant cue is embedded in an object in the scene. The DocVQA panel is a receipt with fields, item rows and totals arranged by document layout. The ChartQA panel is a bar chart, where answering requires comparing bar heights and reading values from the plot structure. The evaluation therefore covers natural scene text, document layouts and chart reasoning. The split sizes used throughout the evaluation chapter are listed in Table 4.1.

¹Images adapted from Wikimedia Commons sources [45–47].

Table 4.1: Dataset split sizes.

Dataset	Train	Val	Test	Total
TextVQA	34,602	5,000	5,734	45,336
DocVQA	39,463	5,349	5,188	50,000
ChartQA-H (human)	7,398	960	1,250	9,608
ChartQA-M (machine)	20,901	960	1,250	23,111
ChartQA (H+M, combined)	28,299	1,920	2,500	32,719

4.3 Training Configuration

Before training, all benchmark samples are converted into a unified **LLaVA-style multimodal instruction format** [3]. Each example is stored as an image plus a short user-assistant dialogue, as shown in Listing 4.1, so TextVQA, DocVQA and ChartQA can share the same data loader, prompt template and answer supervision interface. This format is useful because the model sees every task as the same conditional generation problem: given an image and a natural-language user query, generate the assistant answer. It also separates the visual input path from the textual instruction path while keeping the answer span explicit, which makes it easier to mask prompt tokens, compute supervised loss only on assistant tokens and align teacher logits with the same answer positions during distillation.

```

1 {
2   "image": "sample_000123.jpg",
3   "conversations": [
4     {"from": "human", "value": "<image>\nWhat is the invoice number?"},
5     {"from": "gpt", "value": "4572"}
6   ]
7 }
```

Listing 4.1: Representative LLaVA-style sample used after dataset conversion.

We run all training jobs on a single high-memory **NVIDIA H200** graphics processing unit (GPU) or on comparable single-GPU **Vast.ai** instances, depending on availability. Distillation training uses **DeepSpeed ZeRO Stage 2**, **Brain Floating Point 16 (bf16)** mixed precision, **TensorFloat-32 (TF32)** matrix multiplication and gradient checkpointing to control memory use while keeping the student fully trainable and all teachers frozen [48, 49]. Full optimization and routing parameter values are reported in Table B.1 in Appendix B. The engineering details of teacher-logit storage and remote caching are deferred to Section 4.7 for reproducibility purposes.

4.4 Evaluation Metrics

We report benchmark-specific official metrics for each dataset.

TextVQA. For TextVQA, we report the official VQA Accuracy metric. For a predicted answer $\hat{a}$, the per-question score is

$$s_{\text{VQA}} = \min \left(\frac{\# \text{annotators that provided } \hat{a}}{3}, 1 \right),$$

and the final score is the mean over all questions [4].

DocVQA. We report ANLS, the Average Normalized Levenshtein Similarity. The final ANLS is the average score over all samples [5]:

$$\text{NLS}(a, \hat{a}) = 1 - \frac{d_{\text{Lev}}(a, \hat{a})}{\max(|a|, |\hat{a}|)}, \quad s_{\text{ANLS}} = \begin{cases} \text{NLS}(a, \hat{a}), & \text{NLS} \geq 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

ChartQA. For ChartQA, we report Relaxed Accuracy [6]. A prediction is correct if it matches exactly for text answers. For numeric answers, the score handles the zero-target case explicitly:

$$s_{\text{ChartQA}} = \begin{cases} \mathbf{1}[\hat{y} = y], & y = 0, \\ \mathbf{1}\left[\frac{|\hat{y} - y|}{|y|} \leq 0.05\right], & y \neq 0. \end{cases}$$

4.5 Baselines and Compared Methods

For consistency with Chapter 5, we organize the compared methods into four groups: non-distillation references, same-family distillation baselines, different-family single-teacher baselines and heterogeneous teacher baselines. Before introducing the method-specific baselines, the role of each baseline family is grouped in Table 4.2. The grouping separates student-only references, same-family KD, tokenizer-mismatched single-teacher transfer and heterogeneous teacher methods.

The non-distillation references consist of the original student and the full-finetuning baseline, which provide student-only reference points. The same-family baselines use **SmolVLM2-2.2B-Instruct** as teacher for forward KL, reverse KL and Jensen-Shannon distillation, isolating standard teacher-student transfer when the tokenizer and output space are matched [11, 15, 29, 50, 51]. The different-family single-teacher setting uses ULD with a teacher outside the SmolVLM family, testing tokenizer-mismatched transfer without teacher routing [16]. The heterogeneous teacher setting compares uniform averaging, SHM-CKA, REINFORCE and GRACE over a mixed teacher pool, separating static averaging, hidden-state matching and adaptive teacher assignment [17, 21, 22, 31].

Table 4.2: Baseline groups.

Group	Methods	Teacher setup	Purpose
Non-distillation references	Original, full finetuning	No teacher	Student-only reference points
Same-family baselines	Forward KL, Reverse KL, Jensen-Shannon	SmolVLM2-2.2B-Instruct	Standard single-teacher objectives within the SmolVLM family
Cross-family single-teacher	ULD	Qwen2-VL-2B-Instruct	Cross-family transfer without routed teacher selection
Heterogeneous teacher baselines	Uniform averaging, SHM-CKA, REINFORCE, GRACE (Ours)	Mixed teacher set from multiple VLM families	Static averaging vs. hidden-state matching vs. adaptive teacher assignment

For the trainable competitor methods, we keep the supervised term $\mathcal{L}_{CE}$. The logits-only baselines differ only in the KD signal, but SHM-CKA also adds a hidden-state alignment term and REINFORCE also adds a REINFORCE-style policy term [19, 20, 52]. In the descriptions below, γ denotes a generic KD mixing coefficient. α_{KD} is reserved for the GRACE training objective as defined in Equation 3.32.

Original. Directly evaluates the released student without additional optimization, thereby measuring the native capability of the student model before task adaptation or any teacher-guided distillation signal.

Full finetuning. Optimizes only $\mathcal{L}_{CE}$ and uses no teacher, which isolates the effect of supervised task adaptation from any teacher-guided distillation signal.

Forward KL. Uses $\mathcal{L}_{KD} = \text{KL}(p_t \| p_s)$ for same-family soft-target distillation when teacher and student vocabularies are aligned [11].

Reverse KL. Uses $\mathcal{L}_{KD} = \text{KL}(p_s \| p_t)$ [51, 53] as a student-centered divergence baseline, which tests whether matching the teacher from the student’s current distribution is suitable for VLM distillation. Because the reverse-KL gradient is weighted by p_s , this objective tends to emphasize regions where the student already assigns probability.

Jensen-Shannon. Uses the symmetric divergence

$$\frac{1}{2}\text{KL}(p_s \| m) + \frac{1}{2}\text{KL}(p_t \| m), \quad m = \frac{1}{2}(p_s + p_t).$$

It tests whether a bounded symmetric objective is more stable than KL [29, 51].

ULD. Uses $\mathcal{L}_{KD} = \|\pi(p_s) - \pi(p_t)\|_1$ [16]. This is the main tokenizer-mismatched logits baseline because it compares ranked probability mass, not token IDs.

SHM-CKA. Extends ULD with a hidden-state CKA penalty [16, 19, 31]. For the heterogeneous setting, both the ULD and layer-matching terms are averaged uniformly across teachers before combining with $\mathcal{L}_{CE}$. Full implementation details are given in Appendix B.1.

Uniform multi-teacher. Assigns fixed equal weights to all teachers in the heterogeneous pool, $\mathcal{L}_{KD} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{KD}^{(i)}$ [20, 21], thereby testing whether teacher diversity alone is sufficient without adaptive routing or gradient-aware weighting.

REINFORCE. Replaces deterministic routing with a stochastic policy that samples teachers and updates the selector with REINFORCE [20, 52], providing an adaptive routing baseline that does not use the gradient-agreement refinement introduced by GRACE. The selector is trained with the policy-gradient term

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (4.1)$$

where r is the reward, b is an EMA reward baseline and $\mathbf{m}$ is the sampled teacher mask. Its final training objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (4.2)$$

Additional baseline implementation are reported in Appendix B.1.

4.6 Ablation Settings

The ablation study in Section 5.3 isolates the main components of GRACE on DocVQA while keeping the teacher set and training protocol close to the final method. The variants change one mechanism at a time so that the result table can separate teacher diversity, sparse routing, gradient-agreement weighting and EMA score smoothing. The effect of EMA smoothing is isolated in the $\beta_{\text{EMA}} = 0$ ablation in Section 5.3.

Cross-family ULD. The single-teacher cross-family baseline uses one teacher outside the SmolVLM family and optimizes

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{\text{KD}} \mathcal{L}_{ULD}, \quad \mathcal{L}_{ULD} = \|\pi(p_s) - \pi(p_t)\|_1.$$

Uniform averaging. The uniform variant adds the full heterogeneous teacher pool but assigns fixed equal weights:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha_{\text{KD}} \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{TW}^{(i)}.$$

This setting tests whether teacher diversity alone is enough without adaptive selection.

Routing only. The routing-only variant keeps sparse teacher selection but replaces post-routing weights with uniform weights within the routed set:

$$w_i^{(b)} = u_i^{(b)} = \frac{\mathbf{1}[i \in \mathcal{R}_b]}{|\mathcal{R}_b|}.$$

This setting tests whether top- k routing is useful without gradient-agreement refinement.

Full GRACE. The full method combines trie-Wasserstein KD, sparse routing and gradient-agreement weighting:

$$w_i^{(b)} \propto (\hat{p}_i^{(b)})^{\lambda_{\text{blend}}} (\omega_i^{(b)})^{1-\lambda_{\text{blend}}}.$$

Two additional ablation variants isolate the final weighting mechanism, reported in Table 5.6. The $\lambda_{\text{blend}} = 1.0$ variant keeps the active-set threshold and fallback rules but removes the score-softmax factor from the confident-score case:

$$w_i^{(b)} \propto \hat{p}_i^{(b)} \mathbf{1}[i \in \mathcal{A}_b].$$

This row is an active-set-restricted router-weight ablation, not a pure router-only ablation over all routed teachers. The $\beta_{\text{EMA}} = 0$ variant removes temporal smoothing by setting $\hat{s}_i = s_i$ before score-softmax weighting.

4.7 Implementation Details

This section records the engineering details needed to reproduce the main training configuration. These details are useful for systems reproducibility, but they are separate from the fair-comparison protocol summarized earlier. The full optimization and routing parameters are listed in Table B.1. A typical full H200 run takes about 7 hours and costs about 14.41 USD at 2.059 USD/hour on Vast.ai, excluding the separate cost of offline teacher-logit generation and remote storage.

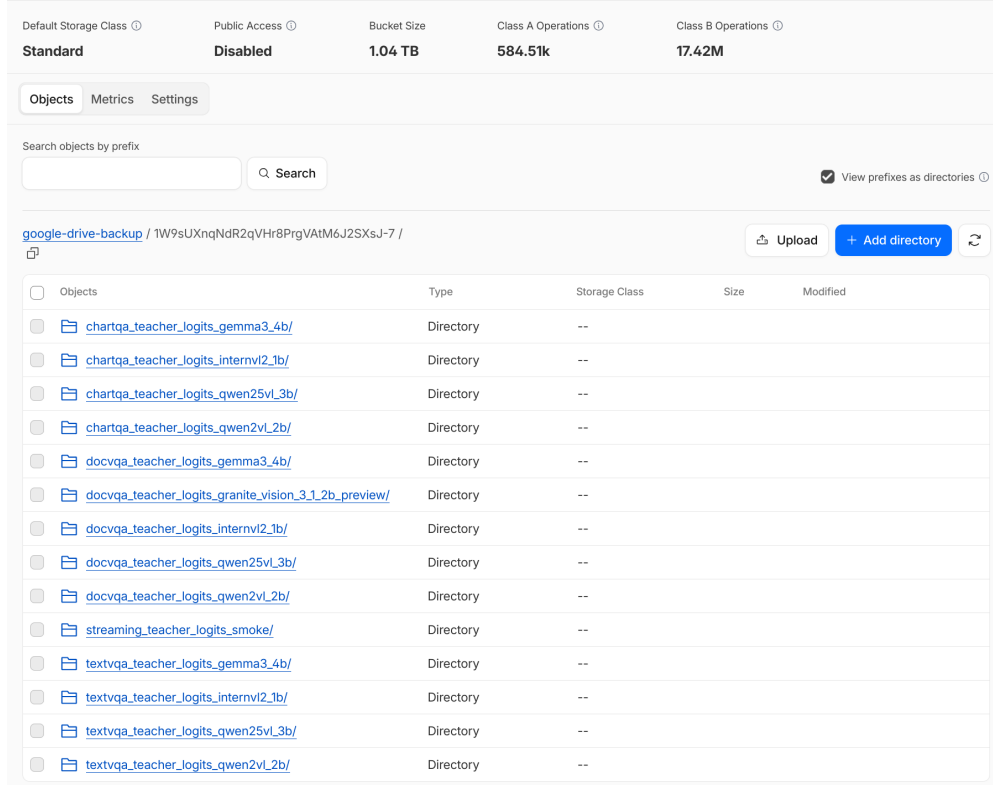


Figure 4.2: R2 teacher-logit storage.

4.7.1 Offline Teacher-Logit Preparation

Teacher logits are generated *offline* before distillation. An offline caching script runs each frozen teacher on every training example, keeps only the supervised answer-token positions and writes one `.pt` file per sample under a teacher-specific directory. Each record stores the sample index, the selected teacher logits and the corresponding supervised labels. This keeps the online training loop simple: the student only reads precomputed teacher targets, so the teachers do not run again. After generation, the teacher-logit directories are uploaded to a **Cloudflare R2** bucket using its S3-compatible object-storage interface [54, 55]. In the active setup, the bucket already exceeds **1 TB**, so copying the full cache to an external GPU instance is impractical. This is especially relevant on services such as Vast.ai, where local NVMe space is usually much smaller than the full teacher logit store. Figure 4.2 shows this offline storage layout and highlights why the training path must stream teacher logits without staging the full cache locally.

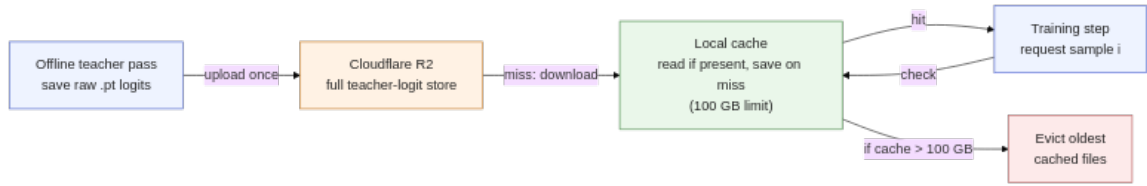


Figure 4.3: Training with remote teacher logits.

4.7.2 Remote Logit Streaming During Training

The main training path does *not* download the full cache before training. Figure 4.3 shows the full remote-cache path: teacher logits are generated offline and uploaded once, each training step requests a sample, the local cache serves hits directly and cache misses trigger an R2 download followed by least-recently-used eviction when the 100 GB limit is exceeded. When the launcher receives an R2 root in `distill_multi_teachers.sh`, the trainer switches to remote lazy loading. During training, `StreamingTeacherLogitsCache` resolves the teacher-specific remote object path for each requested sample, then fetches files lazily from R2 through `s3fs` and `WholeFileCacheFileSystem` [56, 57]. To keep disk usage bounded, the loader maintains a **100 GB** local cache. Each file is cached on first access, its access time is updated on each subsequent open and the oldest files are evicted once the cache exceeds the limit. The local cache therefore behaves as a small least-recently-used buffer over the much larger R2 store. A per-teacher file lock guards cache metadata updates so that concurrent dataloader workers do not corrupt the cache state.

Table 4.3: Logit-streaming settings.

Setting	Value
Teacher-logit storage	Cloudflare R2 object storage
Teacher-logit format	Raw per-sample .pt files
Remote path template	<code>remote_root/<dataset>_teacher_logits_<alias>/<teacher-slug>/<index>.pt</code>
Local cache directory	<code>/tmp/teacher-logits-streaming</code> for R2 reads <code>/workspace/cache</code> for local-cache reads
Local cache limit	100 GB

Chapter 5

Experiment Results

This chapter presents the empirical results for all experiments described in Chapter 4. The main comparison evaluates GRACE against non-distillation references, same-family logits baselines, cross-family single-teacher transfer and heterogeneous multi-teacher methods across TextVQA, DocVQA and ChartQA. The ablation study then isolates the contribution of each component and the efficiency analysis addresses the practical training and storage costs.

5.1 Main Comparison Results

This section reports the main empirical comparison used to answer RQ1. The results are organized from simpler reference points to the full heterogeneous multi-teacher setting: original students, supervised fine-tuning, same-family logits KD, cross-family ULD and multi-teacher distillation methods. The section first establishes the student and teacher reference ranges, then compares all methods on TextVQA, DocVQA and ChartQA. This ordering separates two questions for the reader: whether distillation improves over the student-only reference point and whether adaptive multi-teacher weighting adds value beyond fixed averaging or single-teacher transfer.

5.1.1 Baseline Results

Before applying adaptation or distillation, we report the original performance of the two student models, **SmolVLM-500M** and **SmolVLM-256M**, using **VLM-Eval-Kit** [44] under the evaluation pipeline described in Chapter 4. The student-only starting point for all later fine-tuning and distillation comparisons is given in Table 5.1. The teacher-side reference range is reported in Table 5.2. The gap between these student models and the stronger teachers motivates the distillation experiments that follow.

Table 5.1: Original SmolVLM student results.

Model	TextVQA	DocVQA	ChartQA
SmolVLM-500M	56.75	59.52	62.80
SmolVLM-256M	47.17	48.64	45.04

Table 5.2: Teacher and reference-model results.

Teacher Model	TextVQA	DocVQA	ChartQA
Qwen2.5-VL-3B-Instruct	79.30	93.90	94.24
InternVL3-1B	74.10	81.90	75.30
Gemma-3-4B-it	57.80	75.80	63.60
Qwen2-VL-2B-Instruct	79.70	90.10	89.28

5.1.2 Overall Comparison

The main benchmark results across original models, direct fine-tuning, standard logits distillation baselines and GRACE are collected in Table 5.3. This main result table reports the strongest completed run for each method and benchmark. Uniform, SHM-CKA, REINFORCE and GRACE use a four-teacher mixed set spanning the Qwen, InternVL and Gemma families. Chapter 4 documents the four-teacher configuration used across all multi-teacher experiments. We report the baselines in four groups. The non-distillation references are the original student and full-finetuning runs. The same-family baselines use **SmolVLM2-2.2B-Instruct** as teacher for forward KL, reverse KL and Jensen-Shannon divergence [11, 12, 31]. The different-family single-teacher setting uses ULD with a teacher outside the SmolVLM family [16]. The multi-teacher baselines compare four-teacher Uniform averaging with adaptive methods such as SHM-CKA, REINFORCE and GRACE [20–22, 52].

Table 5.3: Main comparison across TextVQA, DocVQA and ChartQA.

Category	Method	Teacher	Student	TextVQA	DocVQA	ChartQA
Non-distillation references						
Original	None	None	SmolVLM-500M	56.75	59.52	62.80
			SmolVLM-256M	47.17	48.64	45.04
Reference	Finetuning	None	SmolVLM-500M	56.96	61.96	63.00
			SmolVLM-256M	49.83	42.03	49.68
Same-family distillation baselines						
Same-family	Forward KL	SmolVLM2-2.2B	SmolVLM-500M	59.00	63.55	64.52
			SmolVLM-256M	50.88	49.46	56.80
	Reverse KL		SmolVLM-500M	44.73	29.28	30.16
			SmolVLM-256M	29.75	23.28	17.20
	Jensen-Shannon		SmolVLM-500M	42.33	36.43	43.75
			SmolVLM-256M	33.84	33.12	50.93
Different-family single-teacher baselines						
Cross-family	ULD	Qwen2-VL-2B-Instruct	SmolVLM-500M	61.49	65.28	66.92
			SmolVLM-256M	53.95	52.24	60.96
Multi-teacher baselines						
Multi-teacher	Uniform	†	SmolVLM-500M	63.88	69.59	75.60
			SmolVLM-256M	54.57	55.70	63.68
	SHM-CKA		SmolVLM-500M	60.47	63.32	76.88
			SmolVLM-256M	55.93	57.24	62.40
	REINFORCE		SmolVLM-500M	63.26	64.00	70.27
			SmolVLM-256M	56.31	58.53	68.74
	GRACE (Ours)		SmolVLM-500M	69.38	73.11	80.20
			SmolVLM-256M	60.73	60.48	72.32

† Teacher pool: Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct, InternVL3-1B, Gemma-3-4B-it.

On DocVQA, GRACE rises from 69.59 to 73.11 on SmolVLM-500M and from 55.70 to 60.48 on SmolVLM-256M relative to the Uniform baseline. ChartQA shows the largest margins over Uniform: 75.60 to 80.20 for SmolVLM-500M and 63.68 to 72.32 for SmolVLM-256M. Several trends are consistent across the three benchmarks. Same-family logits baselines remain weak: forward KL gives only small gains, reverse KL is consistently poor and Jensen-Shannon is unstable. Cross-family ULD is already stronger than the same-family logits losses on all three tasks, showing that tokenizer-mismatched distillation is more useful here than vocabulary-matched KL. Once multiple teachers are added, Uniform averaging gives another clear gain, which means teacher diversity itself is valuable. The additional improvement from single-teacher ULD to GRACE comes from the full recipe: trie-Wasserstein KD, sample-dependent sparse routing and gradient-agreement refinement after routing. GRACE also improves on the uniform multi-teacher baseline for both student sizes. The gain is moderate on TextVQA and DocVQA, but larger on ChartQA, indicating that adaptive weighting helps most when the task mixes visual reading, numerical reasoning and teacher specialization across datasets.

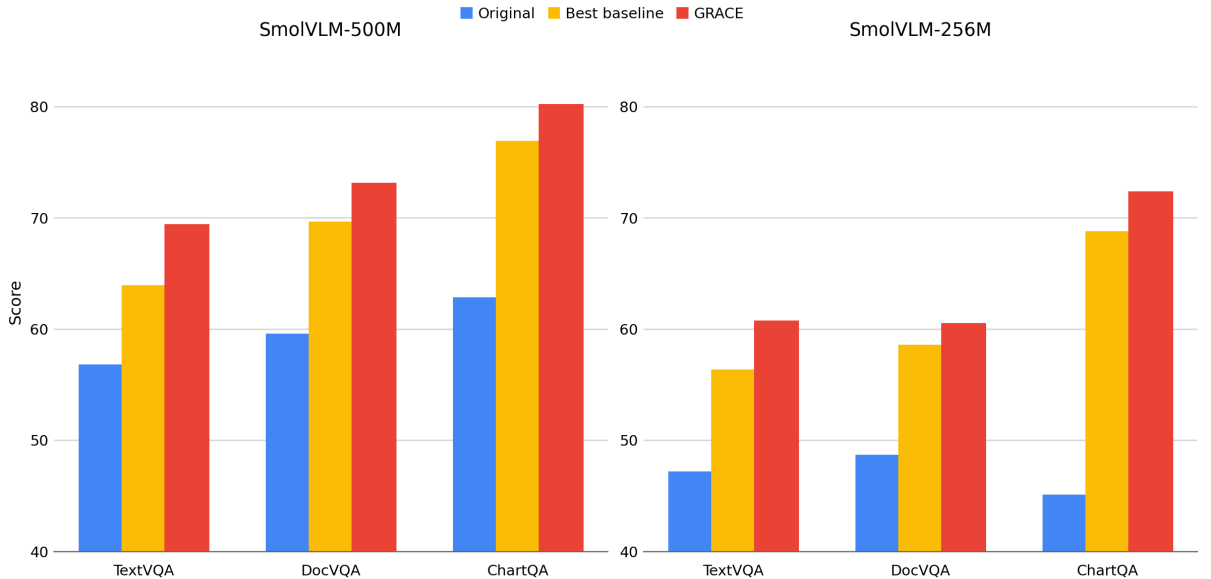


Figure 5.1: Grouped benchmark comparison.

Figure 5.1 summarizes the main ranking by showing that GRACE remains above the strongest non-GRACE baseline across all three benchmarks and both student sizes. To provide qualitative evidence of attention improvement, we use DEX-AR attribution maps [58]. For token t , layer l and head h , DEX-AR forms a gradient-based attribution map

$$\mathbf{G}_t^{(l,h)} = \nabla_{\mathbf{A}^{(l,h)}} y_t \odot \mathbf{A}^{(l,h)},$$

then keeps visually grounded heads with

$$w^{(l,h)} = \max(S_{\text{img}}^{(l,h)} - S_{\text{text}}^{(l,h)}, 0),$$

and aggregates

$$\mathbf{H}_t = \sum_{l,h} w^{(l,h)} \mathbf{G}_t^{(l,h)}, \quad \mathbf{H}_{\text{sent}} = \sum_t \delta_t \mathbf{H}_t.$$

Here $\delta_t = 1/|\Omega_{\text{ans}}|$ for answer tokens $t \in \Omega_{\text{ans}}$, where Ω_{ans} is the set of generated answer token positions and $\delta_t = 0$ otherwise. The sentence-level map is therefore the uniform average of token-level maps over the generated answer span. Raw attention maps and generic saliency methods are not well suited here because the model is autoregressive and the quantity of interest is which image regions drove the answer, not which regions are visually salient in general. DEX-AR is designed for this target: it attributes each answer token back to the input image and averages those attributions across the answer span. Figure 5.2 summarizes the pipeline before the qualitative TextVQA example, making the later heatmaps easier to interpret as answer-grounding evidence, not generic saliency maps.

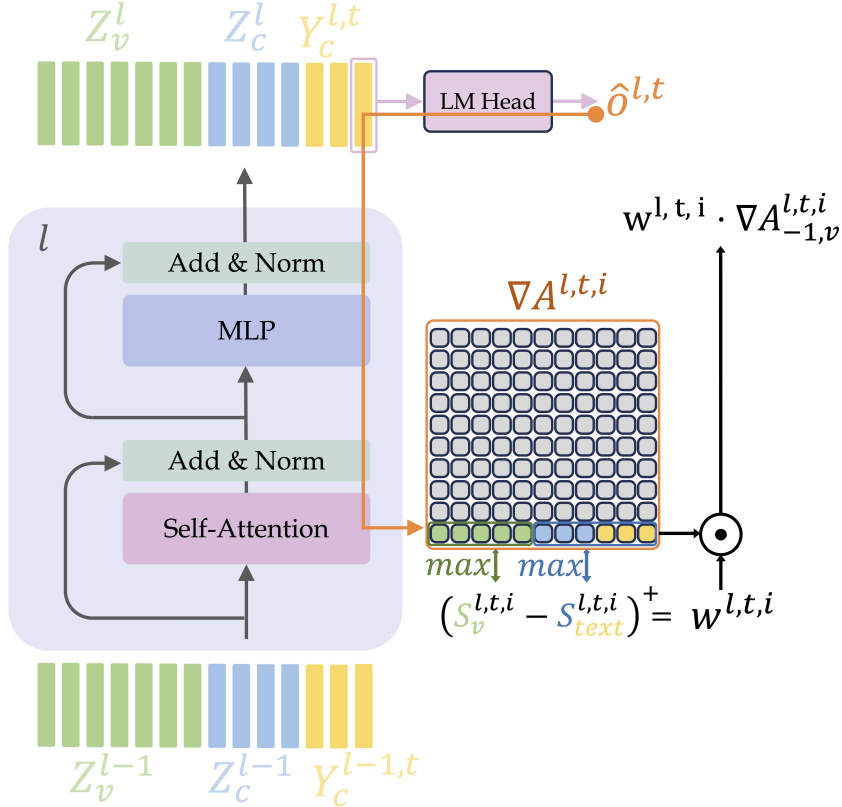


Figure 5.2: DEX-AR overview.

Figure 5.3 gives a representative TextVQA example. Sample 3 asks “*who is this copyrighted by?*” The relevant visual evidence is the copyright text on the lower-left border. The original SmolVLM-500M assigns high attention to the *Virgin* logo, not this region. GRACE shifts attention toward the copyright text, reducing this error.



(a) Input image



(b) Original SmolVLM-500M



(c) GRACE SmolVLM-500M

Figure 5.3: Representative TextVQA attention example.

Answer to RQ1

GRACE improves over direct finetuning, same-family logits KD, ULD and the compared heterogeneous multi-teacher baselines in Table 5.3. The ranking is clear: same-family logits baselines are weakest, cross-family ULD is stronger, multi-teacher baselines improve further and GRACE is best overall. Within the multi-teacher group, static Uniform averaging already helps, but GRACE stays ahead of SHM-CKA, REINFORCE and Uniform itself. This shows that teacher diversity already helps, but the largest gains come from combining the trie-Wasserstein loss with gradient-based alignment. Table 5.5 quantifies how much of the teacher-student gap is closed by GRACE.

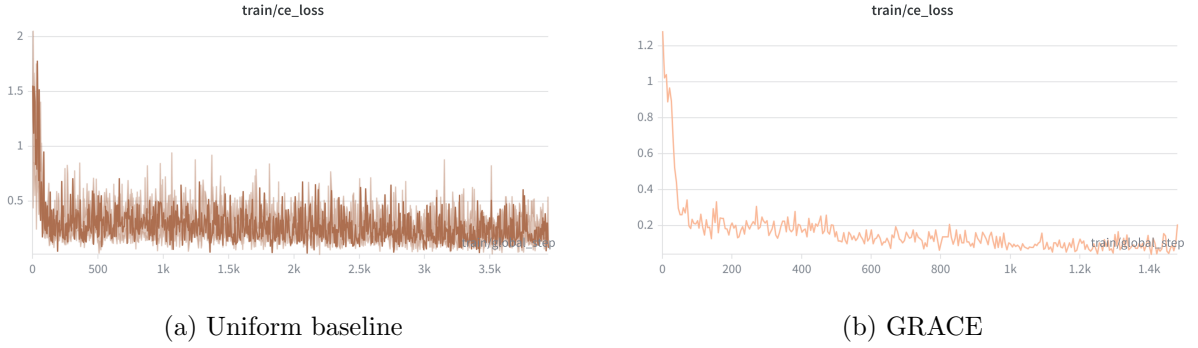


Figure 5.4: Uniform and GRACE training curves.

Figure 5.4 shows that the GRACE run produces a smoother and more stable training curve than Uniform on DocVQA. The Uniform curve shows sharper oscillations, while the GRACE curve decreases more steadily after the early training phase.



(a) Without GRACE refinement



(b) With GRACE

Figure 5.5: Routed-teacher load examples.

Figure 5.5 contrasts collapsed and stable routing behavior. Without the full GRACE recipe, the routed load concentrates on one teacher, reducing teacher diversity. With GRACE, the load stays distributed across the teacher pool, so the student continues to receive heterogeneous supervision. The relative gains in Table 5.4 are consistent with this routing behavior. Relative gain is computed as $(\text{GRACE} - \text{baseline})/\text{baseline}$.

Table 5.4: Relative gains of GRACE.

Benchmark	vs Original	vs Finetuning	vs Uniform	vs Best non-GRACE
<i>SmolVLM-500M</i>				
TextVQA	+12.63 (+22.26%)	+12.42 (+21.80%)	+5.50 (+8.61%)	+5.38 (+8.41%)
DocVQA	+13.59 (+22.83%)	+11.15 (+18.00%)	+3.52 (+5.06%)	+3.52 (+5.06%)
ChartQA	+17.40 (+27.71%)	+17.20 (+27.30%)	+4.60 (+6.08%)	+3.32 (+4.32%)
<i>SmolVLM-256M</i>				
TextVQA	+13.56 (+28.75%)	+10.90 (+21.88%)	+6.16 (+11.29%)	+4.42 (+7.85%)
DocVQA	+11.84 (+24.34%)	+18.45 (+43.90%)	+4.78 (+8.58%)	+1.95 (+3.33%)
ChartQA	+27.28 (+60.57%)	+22.64 (+45.57%)	+8.64 (+13.57%)	+3.58 (+5.21%)

The gains over the original SmolVLM-500M student are large: +22.26% on TextVQA, +22.83% on DocVQA and +27.71% on ChartQA. The gains over direct finetuning are almost as large, at +21.80%, +18.00% and +27.30%, respectively. The gain over static uniform averaging is smaller, but it remains consistent across all three tasks, isolating the value of adaptive teacher weighting from the value of simply using more than one teacher. The same trend holds for SmolVLM-256M, with relative gains of +28.75%, +24.34% and +60.57% on the three benchmarks. For the gap-to-teacher analysis, let the reference point be the strongest teacher in the GRACE comparison pool reported in Table 5.2. Let $g_{\text{orig}} = y_T - y_{\text{orig}}$ and $g_{\text{grace}} = y_T - y_{\text{grace}}$. The fraction of the gap closed is $1 - g_{\text{grace}}/g_{\text{orig}}$.

Table 5.5: Gap-to-teacher analysis for SmolVLM-500M.

Benchmark	Best teacher	Original gap	GRACE gap	Gap closed
TextVQA	79.70	22.95	10.32	55.03%
DocVQA	93.90	34.38	20.79	39.53%
ChartQA	94.24	31.44	14.04	55.34%

The gain is not merely an artifact of comparison with weak baselines. On two benchmarks, GRACE removes about half of the original teacher-student gap and even on DocVQA, where the teacher ceiling is highest, it still closes almost 40% of the gap.

The distilled student can outperform individual teachers. On TextVQA, GRACE reaches 69.38 and exceeds Gemma-3-4B-it at 57.80 by +11.58 points (+20.03%). On ChartQA, it exceeds InternVL3-1B at 75.30 by +4.90 points (+6.51%) and Gemma-3-4B-it at 63.60 by +16.60 points (+26.10%). This shows that routed multi-teacher supervision can combine useful signals that are not available from any single teacher alone.

One negative result is also consistent. Reverse KL is much weaker than forward KL. This can be explained by the different student-logit gradients. Let $p_s = \text{softmax}(z_s)$ be the student distribution and let p_t be the fixed teacher distribution. For forward KL,

$$\mathcal{L}_{FKL} = \text{KL}(p_t \| p_s) = \sum_v p_t(v) \log \frac{p_t(v)}{p_s(v)}, \quad \frac{\partial \mathcal{L}_{FKL}}{\partial z_s(v)} = p_s(v) - p_t(v).$$

Thus, every teacher-supported token directly pulls the student distribution toward the teacher’s soft target [11]. Reverse KL gives

$$\mathcal{L}_{RKL} = \text{KL}(p_s \| p_t) = \sum_v p_s(v) \log \frac{p_s(v)}{p_t(v)}.$$

The student-logit gradient is

$$\frac{\partial \mathcal{L}_{RKL}}{\partial z_s(v)} = p_s(v) \left(\log \frac{p_s(v)}{p_t(v)} - \mathcal{L}_{RKL} \right).$$

The key difference is the factor $p_s(v)$. If $p_t(v) > 0$ but $p_s(v) \rightarrow 0$, then

$$\frac{\partial \mathcal{L}_{FKL}}{\partial z_s(v)} \rightarrow -p_t(v), \quad \frac{\partial \mathcal{L}_{RKL}}{\partial z_s(v)} \rightarrow 0.$$

Teacher-supported tokens that the student currently assigns low probability therefore receive weaker correction under reverse KL. In finite training, this student-weighted gradient can make reverse KL less useful for answer-token distillation, especially when teacher distributions contain useful secondary mass over OCR variants, numeric formats or plausible answer alternatives. We interpret this as an optimization-bias explanation, not as a universal rule, because recent LLM distillation work shows that the simple mode-seeking interpretation of reverse KL does not always fully describe KD behavior [59].

SHM-CKA alone is also not sufficient because representation similarity is not the same as task-useful output supervision. CKA measures hidden-state geometry [19], but hidden-state similarity does not imply output equivalence:

$$\text{CKA}(H_s, H_t) \approx 1 \not\Rightarrow p_s(\cdot | x) \approx p_t(\cdot | x).$$

A simple linear-head example shows why. Suppose the teacher representation is an orthogonal rotation of the student representation, $H_t = H_s R$ with $R^\top R = I$. Linear CKA gives

$$\text{CKA}(H_s, H_t) = \frac{\|H_t^\top H_s\|_F^2}{\|H_t^\top H_t\|_F \|H_s^\top H_s\|_F} = 1.$$

However, the answer logits can still differ:

$$z_s = H_s W_s, \quad z_t = H_t W_t = H_s R W_t.$$

Unless $W_s = RW_t$, the two models can have identical CKA but different logits and different answer-token distributions. This distinction matches the difference between representational similarity and functional similarity discussed in prior work [60]. In heterogeneous VLM distillation, the final answer distribution also depends on the language head, tokenizer, output vocabulary and task-specific teacher behavior. Although SHM-CKA is competitive on ChartQA, it remains weaker than GRACE on TextVQA and DocVQA, where output-level tokenizer-agnostic matching and sample-dependent teacher weighting are more important.

5.2 Representation Similarity Analysis

We use Centered Kernel Alignment (CKA) as an analysis tool to examine whether teacher and student layers exhibit clear correspondence. For centered hidden-state matrices $\tilde{\mathbf{H}}_s$ and $\tilde{\mathbf{H}}_t$, linear CKA is

$$\text{CKA}(\tilde{\mathbf{H}}_s, \tilde{\mathbf{H}}_t) = \frac{\left\| \tilde{\mathbf{H}}_t^\top \tilde{\mathbf{H}}_s \right\|_F^2}{\left\| \tilde{\mathbf{H}}_t^\top \tilde{\mathbf{H}}_t \right\|_F \left\| \tilde{\mathbf{H}}_s^\top \tilde{\mathbf{H}}_s \right\|_F}. \quad (5.1)$$

Values closer to 1 indicate stronger representation similarity [19]. CKA is therefore used here as a representation-analysis tool, not as evidence that two models implement the same answer function. A high CKA score indicates similar hidden geometry on the evaluated samples, but it does not guarantee similar answer-token probabilities after the language head. The main question is whether strong internal similarity is enough to justify layer-wise distillation. The results show that it is not, as the figures and tables below confirm.

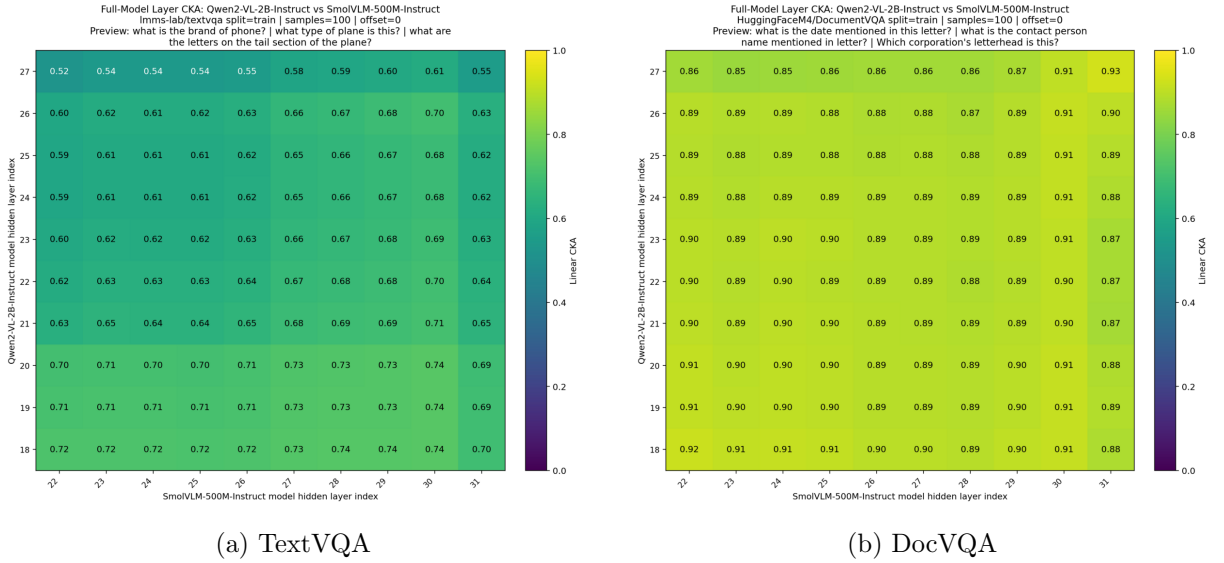


Figure 5.6: Final-layer CKA heatmaps.

Figure 5.6 compares late hidden layers of Qwen2-VL-2B-Instruct and SmolVLM-500M. The TextVQA heatmap contains moderate similarities, mostly around 0.55 to 0.74, with higher values toward later student layers. The DocVQA heatmap is much brighter, with most entries around 0.87 to 0.93. These values show that the final language-side representations are not severely misaligned, especially on document-style inputs.

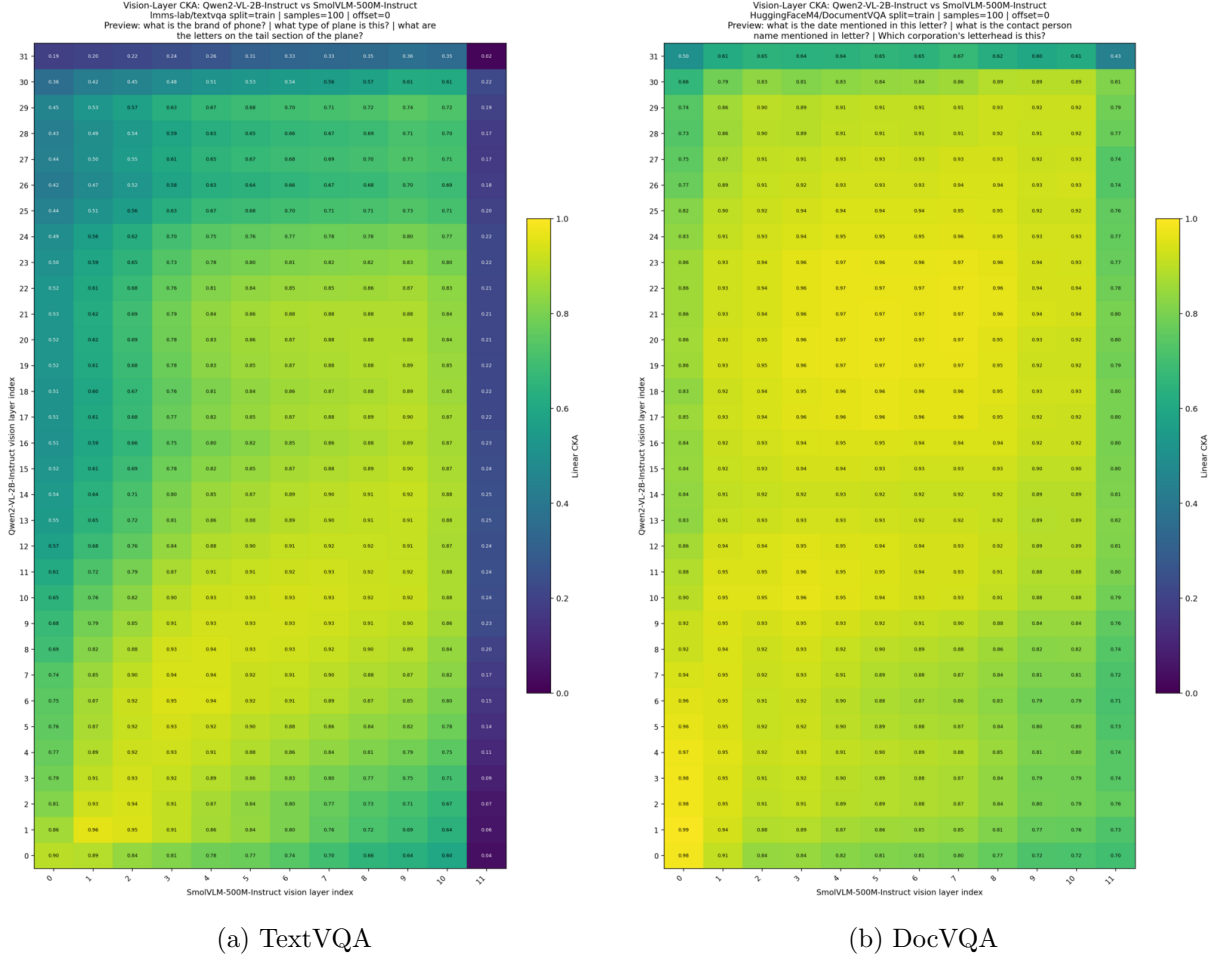


Figure 5.7: Vision-layer CKA heatmaps.

Figure 5.7 shows the same pattern in the vision stack. Both datasets contain broad high-similarity regions and the DocVQA vision map is especially strong across the middle layers. Teacher and student can therefore remain relatively close in both language and vision representations. However, the main benchmark results in Table 5.3 still contain a clear gap between the student models and stronger teachers. The layer-wise baseline also fails to consistently outperform output-level distillation. This pattern supports the design choice in GRACE: output-level tokenizer-agnostic KD and sample-dependent teacher weighting are more reliable than representation matching alone, because high hidden-state similarity is not a reliable signal that direct layer matching will produce better task transfer.

5.3 Ablation Study

This section answers RQ2 from Chapter 1 by isolating the main components of GRACE on DocVQA. Section 4.6 defines the ablation variants, with the corresponding results for both student models reported in Table 5.6.

Table 5.6: DocVQA component ablation.

DocVQA ablation variant	SmolVLM	
	256M	500M
Cross-family ULD (single teacher)	52.24	65.28
Uniform averaging	55.70	69.59
Routing only + trie-Wasserstein	57.45	63.23
Full GRACE + trie-Wasserstein	60.48	73.11
w/o EMA ($\beta_{\text{EMA}} = 0$)	57.45	71.25
w/ fixed blend ($\lambda_{\text{blend}} = 1.0$)	57.97	70.45

Teacher diversity is necessary but not sufficient because uniform multi-teacher averaging adds useful supervision while still failing to resolve tokenizer mismatch, teacher specialization and conflicting KD directions. Across the ablation results in Table 5.6, the full method is the strongest setting for both student sizes. Moving from a single cross-family teacher (ULD) to uniform four-teacher averaging adds +3.46 on SmolVLM-256M and +4.31 on SmolVLM-500M. The gain confirms that multiple heterogeneous teachers provide better supervision than any single teacher, but uniform weighting leaves a further gap of +4.78/+3.52 to the full method.

Routing without gradient refinement can hurt. On SmolVLM-500M, routing-only scoring (63.23) is *worse* than uniform averaging (69.59) by -6.36 points. The router selects teachers but assigns them incorrect relative weights. Without GRACE, these wrong weights degrade the result below a naive equal average. On SmolVLM-256M the pattern reverses: routing-only (57.45) beats uniform (55.70) by +1.75. SmolVLM-500M, the larger student, produces a sharper routing distribution that amplifies weighting errors when uncorrected. SmolVLM-256M produces a flatter routing signal closer to uniform, so routing-only is less damaging and even slightly beneficial.

Gradient-agreement refinement acts as the critical corrective layer because it adjusts routed teacher weights according to whether each teacher’s KD direction agrees with the supervised task objective. Full GRACE recovers +9.88 points over routing-only on SmolVLM-500M and +3.03 on SmolVLM-256M. Decomposing the 500M gain, the active-set router-weight variant ($\lambda_{\text{blend}} = 1.0$) already recovers +7.22 points (63.23 $\rightarrow$ 70.45) by replacing uniform post-routing weights with learned router weights while retaining the GRACE active-set filter. Full score-softmax blending then adds another +2.66 points (70.45 $\rightarrow$ 73.11). On SmolVLM-256M, the active-set router-weight variant adds only +0.52 points (57.45 $\rightarrow$ 57.97), while full score-based blending adds +2.51 points (57.97 $\rightarrow$ 60.48). These results show that active-set filtering and router weighting help, but agreement-score weighting is still needed for the strongest final performance. Removing EMA smoothing ($\beta_{\text{EMA}} = 0$) degrades both students (256M: 60.48 $\rightarrow$ 57.45, 500M: 73.11 $\rightarrow$ 71.25), confirming that temporal smoothing stabilizes score estimates but is secondary to the refinement step itself.

Answer to RQ2

The ablation isolates three cumulative contributions: (1) heterogeneous teacher diversity (+3 to +4 pts over single-teacher ULD), (2) sparse routing structure (beneficial only when paired with refinement) and (3) gradient-agreement refinement, which adds +3 to +10 pts over routing-only and is the single largest contributor. EMA smoothing provides additional stabilization by reducing the effect of noisy batch-level gradient-agreement scores before they are converted into final KD weights. Without gradient refinement, routing can degrade performance below uniform averaging.

5.4 Efficiency Analysis

This section addresses RQ3 from Chapter 1. Runtime, training cost and storage overhead are analyzed using the H200 rental rate and the observed Cloudflare R2 storage footprint.

Answer to RQ3

GRACE is mainly limited by GPU training and offline logit generation, not object storage. Figure 4.2 shows the R2 bucket used for teacher-logit caches, with dataset- and teacher-specific directories and a total footprint of 1.04 TB. A 7-hour H200 run costs about 14.41 USD, while this R2 logit store costs about 15.60 USD/month for storage plus 3.71 USD for billable Class B operations or about 19.31 USD/month before rounding. The practical efficiency bottleneck is compute, not storage.

5.5 Discussion

The results show a clear ordering across method families. Same-family KL baselines are weakest, single cross-family distillation is stronger, uniform multi-teacher weighting improves further and GRACE is best overall [11, 16]. No single component fully addresses the heterogeneous distillation problem: heterogeneous teachers provide better supervision, but the student still needs a tokenizer-agnostic loss, sample-dependent routing and a stable post-routing weighting scheme in the evaluated setting. The training recipe explains this pattern because GRACE combines a shared-trie Wasserstein loss for mismatched tokenizers with routed multi-teacher selection, top- k masking, routed-weight renormalization and EMA-smoothed gradient-agreement refinement [16, 32, 36]. The smoother DocVQA training curve in Figure 5.4 is consistent with that design: the router narrows the teacher set first, then GRACE reweights only the routed teachers, reducing the instability caused by uniformly mixing heterogeneous teacher signals.

First-order interpretation of gradient-agreement refinement. The gradient-agreement component can also be understood from a first-order multi-objective view. Let $g_{CE} = \nabla_{\theta} \mathcal{L}_{CE}$ and $g_i = \nabla_{\theta} \mathcal{L}_{KD}^{(i)}$ be the supervised and teacher-specific KD gradients. A combined update with teacher weights w_i is

$$\Delta\theta = -\eta \left(g_{CE} + \alpha_{KD} \sum_i w_i g_i \right).$$

The first-order change in the supervised loss is

$$\Delta \mathcal{L}_{CE} \approx g_{CE}^{\top} \Delta\theta = -\eta \|g_{CE}\|_2^2 - \eta \alpha_{KD} \sum_i w_i \langle g_{CE}, g_i \rangle.$$

If $\langle g_{CE}, g_i \rangle < 0$, then teacher i contributes a term that increases the first-order CE loss when the supervised objective should decrease. This is the gradient-conflict case studied in multi-task optimization [41]. GRACE does not project gradients as PCGrad does. It uses the cosine agreement

$$\cos(g_{CE}, g_i) = \frac{\langle g_{CE}, g_i \rangle}{\|g_{CE}\|_2 \|g_i\|_2}$$

as a teacher-weighting signal. This explains why post-routing gradient refinement can improve over routing alone. The largest gain appears on ChartQA and the trend holds for SmolVLM-256M as well. Sample-dependent teacher assignment is most useful when the task mixes visual parsing and numerical reasoning and the method benefits both evaluated student models. The comparison with SHM-CKA reinforces this conclusion by showing that heterogeneous teacher access is not enough when the method relies only on hidden-state matching with no routed output-level KD. SHM-CKA is competitive on ChartQA yet clearly weaker on TextVQA and DocVQA. Together with the CKA heatmaps, this confirms that internal similarity alone does not guarantee effective transfer. **Sample-dependent teacher weighting therefore remains necessary, particularly when the teacher pool contains models with different tokenizers, architectures and task-specific strengths [19, 31].**

5.6 Limitations and Threats to Validity

Cost-limited evaluation. Because the reported numbers are single-run results, the comparison does not provide confidence intervals or repeated-seed estimates for small gaps between strong baselines. As noted in Section 5.4, a typical run costs approximately 14.41 USD before offline cache generation and storage cost. Although the main tables use a shared four-teacher pool and consistent hyperparameters, the absence of repeated seeds limits the strength of strict uncertainty claims about closely matched methods in this study. **Scope of validation.** The empirical study focuses on VQA benchmarks and SmolVLM-family students. Although the reported gains are strong in this regime, the thesis does not yet establish whether the same behavior holds for other multimodal task families, other student architectures or larger teacher pools.

Conclusion and Future Work

This thesis studied vision-language model distillation under heterogeneous teachers, where teacher utility varies across samples and teacher-student tokenizers do not match. To address these challenges, we presented Gradient-Routed Alignment with Capacity-aware Experts (GRACE), a routed multi-teacher framework with trie-Wasserstein tokenizer-agnostic distillation, sparse top- k selection and gradient-agreement refinement. The training procedure routes teachers first, refines their weights with gradient-agreement scores and then aggregates the routed trie-Wasserstein KD loss [16, 32, 36]. Empirically, GRACE obtains the highest scores among the methods compared in the main tables, confirming that selective routing and post-routing gradient refinement improve heterogeneous teacher-to-student VLM distillation under the evaluated VQA setting. For SmolVLM-500M, it reaches 69.38 on TextVQA, 73.11 on DocVQA and 80.20 on ChartQA. For SmolVLM-256M, GRACE reaches 60.73 on TextVQA, 60.48 on DocVQA and 72.32 on ChartQA, showing that the method improves both evaluated student models. These results exceed the original students, direct finetuning and all reported distillation baselines in Chapter 5. On SmolVLM-500M, GRACE closes about 55% of the original teacher-student gap on TextVQA and ChartQA and about 40% on DocVQA.

Future Work

Future work should investigate three directions that would strengthen the empirical scope, reduce systems cost and extend the method beyond answer-token distillation. First, **stronger evaluation** would use repeated runs with multiple seeds to tighten the comparison, especially for the smaller gaps between close multi-teacher baselines. Second, **systems efficiency** would reduce storage and I/O cost, since the pipeline still relies on large offline teacher-logit caches. This would make the method easier to scale to more datasets and larger teacher sets. Third, **broader method extensions** would move beyond answer-token-centric distillation through visual-token alignment, cross-attention transfer or region-level matching. Future work can also explore richer router inputs, improved objective-conflict handling and teacher sets that mix specialized document, chart and general VLM experts [20–22].

Appendix A

Closed-Form Trie-Wasserstein Derivation

This appendix derives the closed-form expression used by the KD loss in Chapter 3. The result is stated for a general rooted tree [17, 32].

A.1 Setup

Let $T = (V, E, r)$ be a rooted tree with root r . Each edge $e \in E$ has a positive weight $w(e) > 0$. For an edge $e = (p(e), c(e))$, where $p(e)$ is the parent and $c(e)$ is the child, let

$$\Gamma(e) = \{v \in V \mid e \text{ lies on the path from } r \text{ to } v\}$$

denote the subtree below edge e . For two vertices $u, v \in V$, define the tree metric

$$d_T(u, v) = \sum_{e \in \text{path}(u, v)} w(e),$$

where $\text{path}(u, v)$ is the unique set of edges on the path between u and v . Let μ and ν be two probability measures on V . In the trie-Wasserstein setting, these correspond to student and teacher token distributions placed on the leaves of the shared byte trie. Measures on leaves can be viewed as measures on all vertices by assigning zero mass to non-leaf vertices. The Kantorovich form of the W_1 optimal transport distance under the tree metric is [32]

$$W_{1,T}(\mu, \nu) = \min_{\pi \in \Pi(\mu, \nu)} \sum_{u \in V} \sum_{v \in V} \pi(u, v) d_T(u, v),$$

where $\Pi(\mu, \nu)$ is the set of couplings satisfying

$$\sum_{v \in V} \pi(u, v) = \mu(u), \quad \sum_{u \in V} \pi(u, v) = \nu(v), \quad \pi(u, v) \geq 0.$$

For each edge e , define the subtree-mass imbalance

$$\Delta(e) = \mu(\Gamma(e)) - \nu(\Gamma(e)).$$

Hence, the closed-form expression for the tree Wasserstein distance is

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\Delta(e)|.$$

A.2 Lower Bound

Consider any feasible transport plan $\pi \in \Pi(\mu, \nu)$. Removing an edge e separates the tree into two components: the subtree $\Gamma(e)$ and its complement $V \setminus \Gamma(e)$. The total source mass initially inside $\Gamma(e)$ is $\mu(\Gamma(e))$, while the total target mass required inside $\Gamma(e)$ is $\nu(\Gamma(e))$. If

$$\mu(\Gamma(e)) > \nu(\Gamma(e)),$$

then at least $\mu(\Gamma(e)) - \nu(\Gamma(e))$ mass must leave $\Gamma(e)$. If

$$\mu(\Gamma(e)) < \nu(\Gamma(e)),$$

then at least $\nu(\Gamma(e)) - \mu(\Gamma(e))$ mass must enter $\Gamma(e)$. Therefore, any feasible plan must move at least

$$|\Delta(e)| = |\mu(\Gamma(e)) - \nu(\Gamma(e))|$$

mass across edge e . Let $q_\pi(e)$ be the amount of mass transported by π whose path crosses edge e . The argument above implies

$$q_\pi(e) \geq |\Delta(e)|.$$

Because the cost of moving mass from u to v is the sum of edge weights along the unique path between them, the total cost of π can be decomposed edgewise:

$$\sum_{u,v \in V} \pi(u,v) d_T(u,v) = \sum_{e \in E} w(e) q_\pi(e).$$

Using $q_\pi(e) \geq |\Delta(e)|$, we obtain

$$\sum_{u,v \in V} \pi(u,v) d_T(u,v) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

Since this holds for every feasible transport plan π , it also holds for the optimal plan:

$$W_{1,T}(\mu, \nu) \geq \sum_{e \in E} w(e) |\Delta(e)|.$$

A.3 Achievability

It remains to show that the lower bound can be achieved. We define a signed measure

$$\sigma(v) = \mu(v) - \nu(v).$$

For each edge e , the subtree imbalance is

$$\Delta(e) = \sum_{v \in \Gamma(e)} \sigma(v).$$

Interpret $\Delta(e)$ as the net amount of mass that must cross edge e . If $\Delta(e) > 0$, the subtree below e contains surplus mass and $\Delta(e)$ mass must move from $\Gamma(e)$ to its complement. If $\Delta(e) < 0$, the subtree has a deficit and $|\Delta(e)|$ mass must enter $\Gamma(e)$. Because T is a tree, there is a unique path between any two vertices. A bottom-up transport construction can move surplus mass upward from subtrees with $\Delta(e) > 0$ and move mass downward into subtrees with $\Delta(e) < 0$. This construction sends exactly $|\Delta(e)|$ mass across each edge e . It satisfies all supply and demand constraints because the total signed mass is zero:

$$\sum_{v \in V} \sigma(v) = \sum_{v \in V} \mu(v) - \sum_{v \in V} \nu(v) = 1 - 1 = 0.$$

The cost of this construction is therefore

$$\sum_{e \in E} w(e) |\Delta(e)|.$$

Thus there exists a feasible transport plan with cost equal to the lower bound. Hence

$$W_{1,T}(\mu, \nu) \leq \sum_{e \in E} w(e) |\Delta(e)|.$$

Combining this upper bound with the lower bound gives

$$W_{1,T}(\mu, \nu) = \sum_{e \in E} w(e) |\mu(\Gamma(e)) - \nu(\Gamma(e))|.$$

Appendix B

Experiment Parameters

This appendix lists the optimization and routing hyperparameters used across all distillation experiments. These settings correspond to the evaluation protocol described in Chapter 4.

Table B.1: Experiment parameters.

Setting	Value
<i>Model setup</i>	
Student model	HuggingFaceTB/SmolVLM-500M-Instruct
Teacher models	Qwen/Qwen2.5-VL-3B-Instruct, Qwen/Qwen2-VL-2B-Instruct, OpenGVLab/InternVL3-1B, google/gemma-3-4b-it
Teacher weighting strategy	routing (GRACE path)
Distillation loss	Trie-Wasserstein
Trie-Wasserstein rho	0.7
Trie-Wasserstein top- k	64
Validation split	validation
<i>Hardware and infrastructure</i>	
Hardware and runtime	Single-GPU H200 or Vast.ai runs
Distributed training	DeepSpeed ZeRO-2
Numerical precision	bf16 enabled, half precision (fp16) disabled, TF32 enabled
FlashAttention-2	Enabled
Training data format	LLaVA-style multimodal instruction JSON
<i>Training</i>	
Max sequence length	16384
Training epochs	2
Per-device batch size	80
Gradient accumulation steps	1
Effective batch size	80
Gradient checkpointing	Enabled
Frozen modules	None
Dataloader workers	4
Logging and checkpointing	Log every 5 steps and save every 150 steps
Seed	42

Continued on the next page.

Table B.1 continued from the previous page.

Setting	Value
<i>Optimizer</i>	
Optimizer	adamw_torch
Adam betas	(0.9, 0.98)
Adam epsilon	1×10^{-7}
Weight decay	0.0
Base learning rate	1×10^{-5}
Vision learning rate	2×10^{-6}
Connector learning rate	1×10^{-5}
Learning-rate schedule	Cosine decay with warmup ratio 0.03
<i>GRACE hyperparameters</i>	
Student/teacher temperature	$\tau_s = \tau_t = 1.0$
KD weight	$\alpha_{\text{KD}} = 0.5$
Top- k routing	$k_{\text{route}} = 2$, with top-scoring teacher masking and selected softmax-weight renormalization
Gate entropy weight	$\lambda_{\text{ent}} = 10^{-3}$
Router z -loss weight	$\lambda_z = 10^{-3}$
GRACE warmup ratio	$\rho_g = 0.05$
GRACE softmax beta	$\beta_{\text{score}} = 4.0$
GRACE blend exponent	$\lambda_{\text{blend}} = 0.9$
GRACE EMA decay	$\beta_{\text{EMA}} = 0.8$
Score-spread fallback threshold	$\epsilon_{\text{fb}} = 0.01$
Alignment threshold	$\delta = 0.15$

B.1 Baseline Implementation Details

This section records the additional baseline definitions used for the comparison methods in Chapter 4.

SHM-CKA. The baseline first loads an offline CKA matrix and, for each student layer s , selects the top- k teacher layers $\mathcal{T}_s$. Their CKA scores are normalized into soft weights $\alpha_{s,t}$. The per-teacher layer loss is

$$\mathcal{L}_{\text{layer}}^{(i)} = \frac{1}{|S|} \sum_{s \in S} \sum_{t \in \mathcal{T}_s} \alpha_{s,t} (1 - \sqrt{\text{CKA}(\mathbf{H}_s, \mathbf{H}_t)}). \quad (\text{B.1})$$

Each teacher also contributes a ULD logits loss $\mathcal{L}_{\text{ULD}}^{(i)}$. For the heterogeneous baseline, the logits and layer-matching terms are averaged across teachers:

$$\bar{\mathcal{L}}_{\text{ULD}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{ULD}}^{(i)}, \quad \bar{\mathcal{L}}_{\text{layer}} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_{\text{layer}}^{(i)}. \quad (\text{B.2})$$

The final SHM-CKA objective is

$$\mathcal{L}_{\text{SHM-CKA}} = \frac{1}{3} (\mathcal{L}_{CE} + \bar{\mathcal{L}}_{\text{ULD}} + \bar{\mathcal{L}}_{\text{layer}}). \quad (\text{B.3})$$

REINFORCE. The policy uses the pooled student hidden state together with per-teacher summary features. A linear policy predicts independent Bernoulli probabilities

$$\boldsymbol{\pi} = \sigma(W \text{LN}(\mathbf{z}) + b), \quad (\text{B.4})$$

then samples a teacher mask $m_i \sim \text{Bernoulli}(\pi_i)$. If no teacher is selected, the highest-probability teacher is used as fallback. The selected mask is normalized into teacher weights

$$w_i = \frac{m_i}{\sum_{j=1}^M m_j}, \quad \mathcal{L}_{KD} = \sum_{i=1}^M w_i \mathcal{L}_{KD}^{(i)}. \quad (\text{B.5})$$

The selector is trained with the policy-gradient term

$$\mathcal{L}_{\text{pol}} = -(r - b) \log p(\mathbf{m} \mid \boldsymbol{\pi}), \quad (\text{B.6})$$

where r is the reward and b is an EMA reward baseline. The total reinforced-selection objective is

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{CE} + \gamma \mathcal{L}_{KD} + \lambda_{\pi} \mathcal{L}_{\text{pol}}. \quad (\text{B.7})$$

References

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